Week 3

Web Programming

CSS Colors

Adding color to our websites

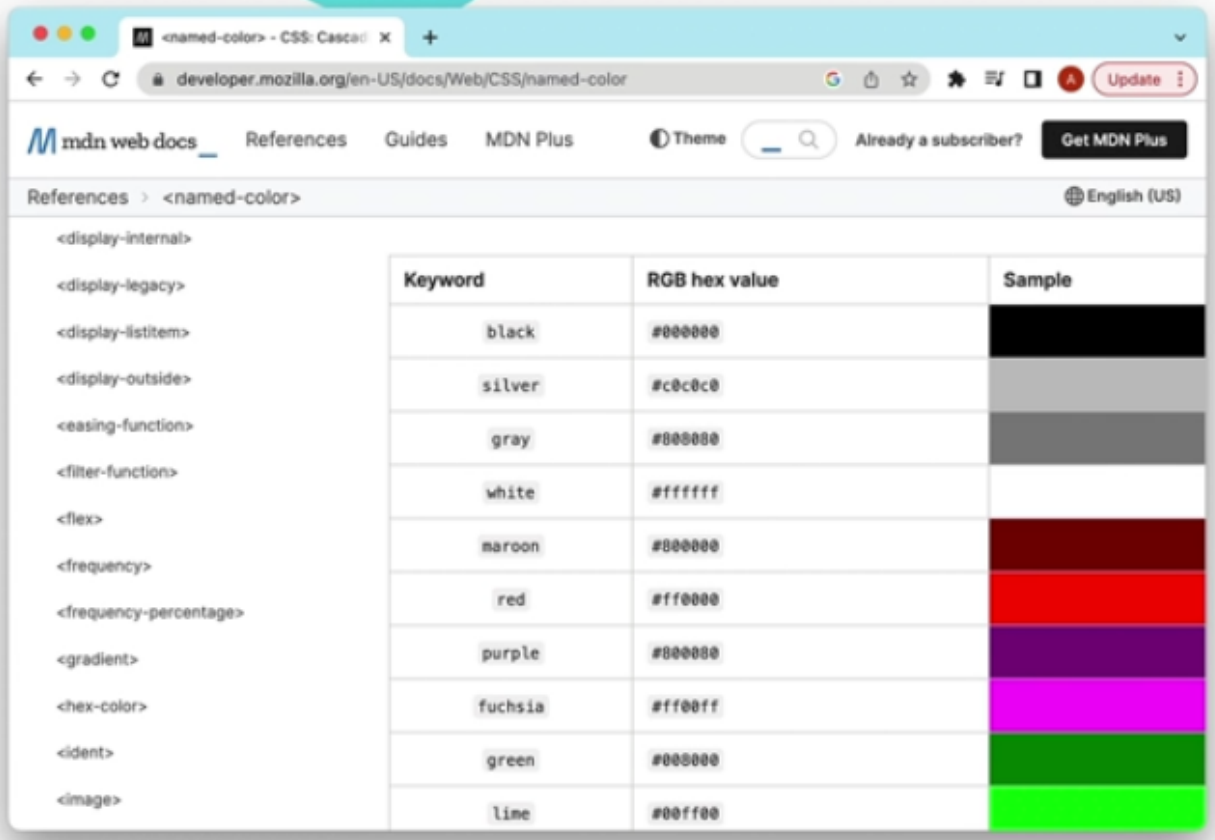


Color Properties



Named Colors

#3CB371 MediumSeaGreen	#40E0D0 Turquoise	#4169E1 RoyalBlue
#4682B4 SteelBlue	#483D8B DarkSlateBlue	
#4B0082 Indigo	#556B2F DarkOliveGreen	
#6495ED CornflowerBlue	#663399 RebeccaPurple	
#696969 DimGray	#696969 DimGrey	
#6B8E23 OliveDrab	#708090 SlateGray	



<frequency-percentage>
<gradient>
<hex-color>
<ident>
<image>
<integer>
<length>
<length-percentage>
<named-color>
<number>
<percentage>
<position>
<ratio>
<resolution>
<shape>
<string>
<system-color>
<time>
<time-percentage>

Syntax

```
color: red;
color: orange;
color: tan;
color: rebeccapurple;
color: transparent;
```

Value

CSS Level 1 values

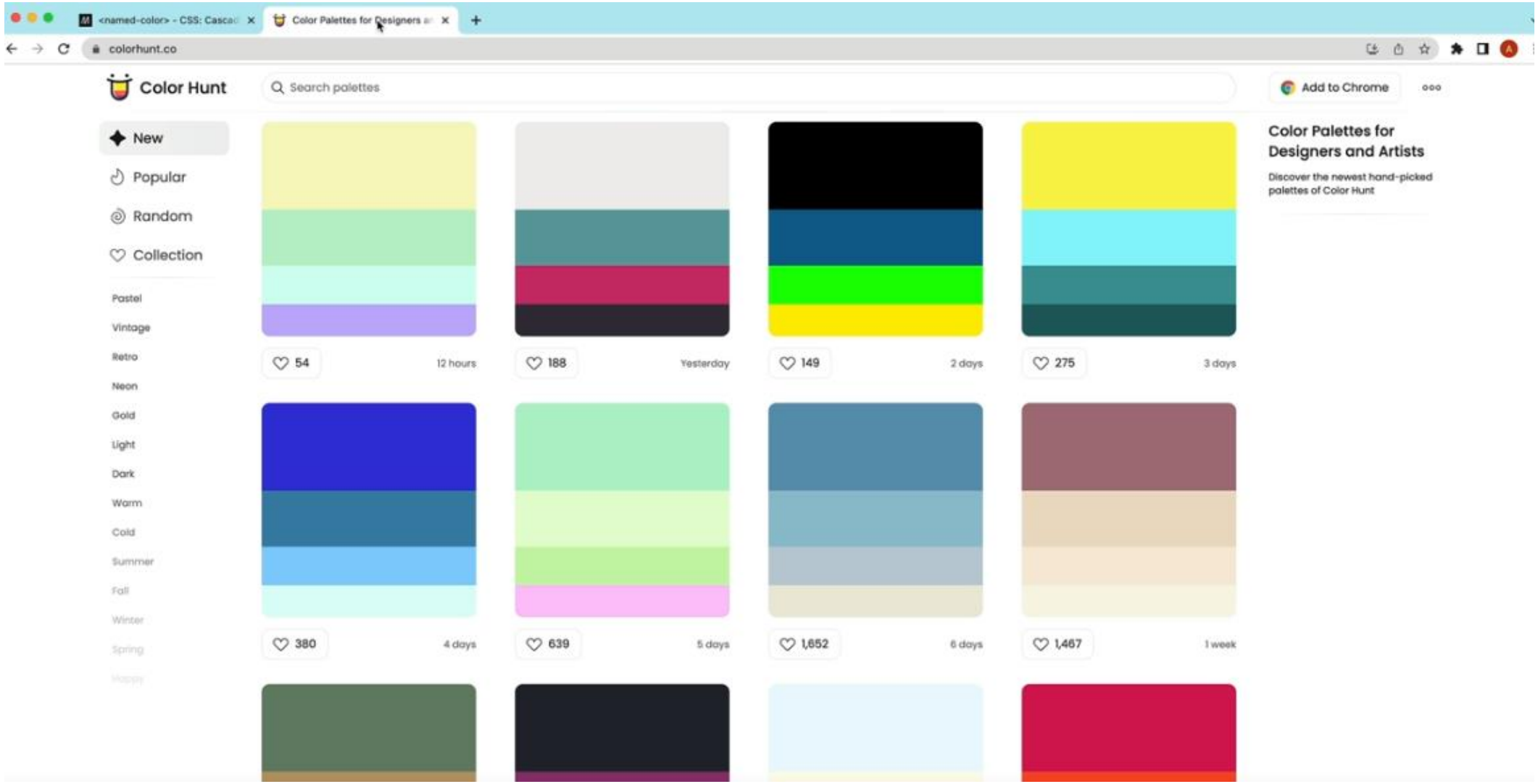
CSS Level 1 only included 16 basic colors, called the *VGA colors* as they were taken from the set of displayable colors on VGA graphics cards.

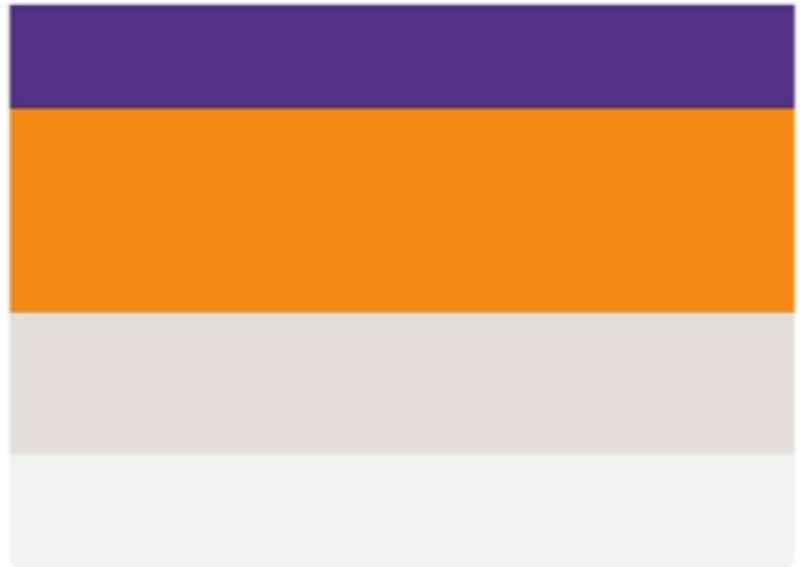
Keyword	RGB hex value	Sample
black	#000000	
silver	#c0c0c0	
gray	#808080	
white	#ffffff	
maroon	#800000	
red	#ff0000	

In this article
Syntax
Description
Examples
Specifications
Browser compatibility
See also

https://developer.mozilla.org/en-US/docs/Web/CSS/named-color

colorhunt.co



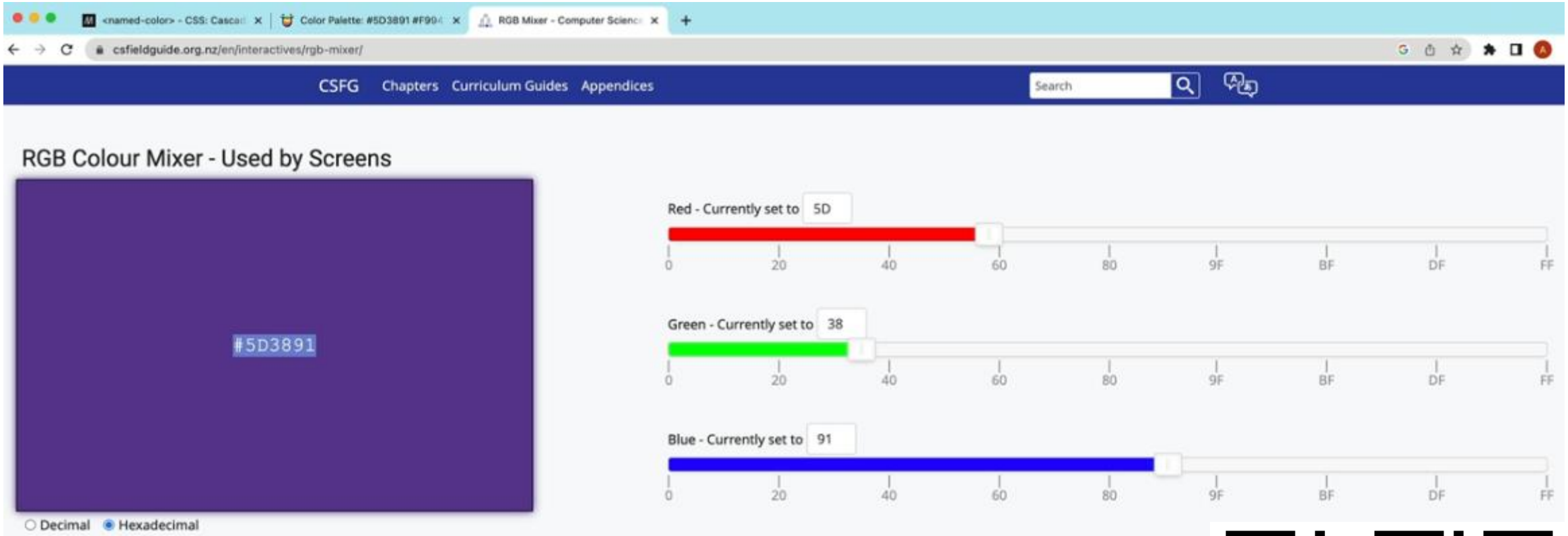


691 Image Link 1 week

Like this palette?

#5D3891	#F99417	#E8E2E2	#F5F5F5
rgb(93, 56, 145)	rgb(249, 148, 23)	rgb(232, 226, 226)	rgb(245, 245, 245)

Purple Orange Grey White Retro Wedding



https://www.csfieldguide.org.nz/en/interactives/rgb-mixer/

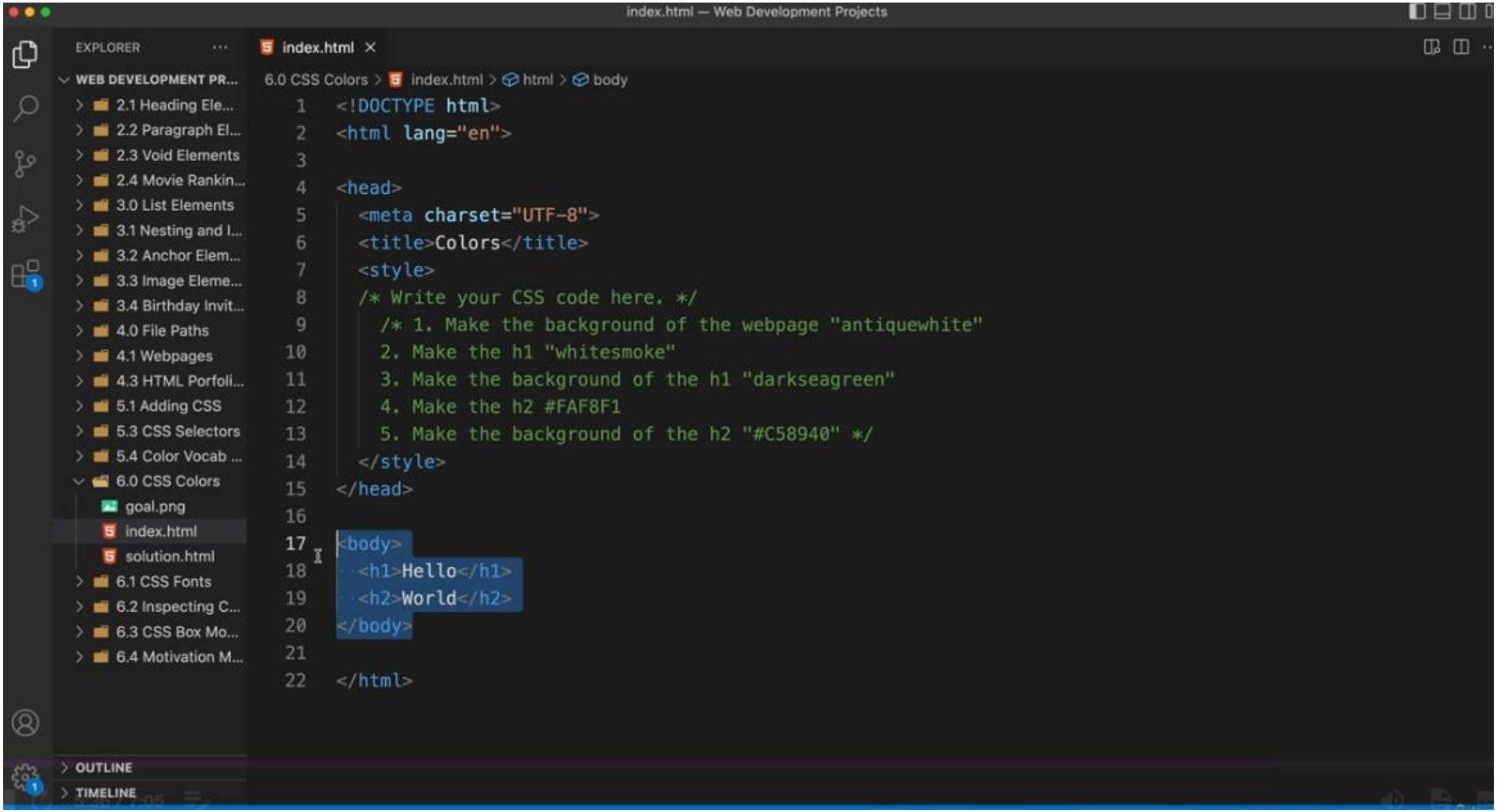
Hex Codes

```
h1 {
  color: #5D3891
}
```

Try an Exercise



6.0 CSS
Colors.zip


index.html — Web Development Projects
EXPLORER
index.html
WEB DEVELOPMENT PR...
2.1 Heading Ele...
2.2 Paragraph El...
2.3 Void Elements
2.4 Movie Rankin...
3.0 List Elements
3.1 Nesting and I...
3.2 Anchor Elem...
3.3 Image Eleme...
3.4 Birthday Invit...
4.0 File Paths
4.1 Webpages
4.3 HTML Porfoli...
5.1 Adding CSS
5.3 CSS Selectors
5.4 Color Vocab ...
6.0 CSS Colors
goal.png
index.html
solution.html
6.1 CSS Fonts
6.2 Inspecting C...
6.3 CSS Box Mo...
6.4 Motivation M...
OUTLINE
TIMELINE
6.0 CSS Colors > index.html > html > body
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5 <meta charset="UTF-8">
6 <title>Colors</title>
7 <style>
8 /* Write your CSS code here. */
9 /* 1. Make the background of the webpage "antiquewhite"
10 2. Make the h1 "whitesmoke"
11 3. Make the background of the h1 "darkseagreen"
12 4. Make the h2 #FAF8F1
13 5. Make the background of the h2 "#C58940" */
14 </style>
15 </head>
16
17 <body>
18 <h1>Hello</h1>
19 <h2>World</h2>
20 </body>
21
22 </html>
Ln 17, Col 1
Spaces: 2
UTF-8
LF
HTML
Go Live

Font Properties

Changing the appearance of text



Font Properties

```
h1 {
  color: blue
  font-weight: bold
  font-size: 20px
  font-family: sans-serif
}
```

Font Size

```
h1 {
  font-size: 20px
}
```

1px

1/96th inch

0.26mm.

1pt

1/72nd inch

0.35mm.

1em

100% of parent

1rem

100% of root

1em

100% of parent

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Document</title>
  <link rel="stylesheet" href="./style.css" />
</head>

<body>
  <h1>Hello</h1>
  <footer>
    <h2>Hello</h2>
  </footer>
</body>

</html>
```

1rem

100% of root

index.html

```html
<html lang="en">
  <body>
    <p id="pixel">1 Pixel is 1/96 of an Inch</p>
    <p id="point">1 Point is 1/72 of an Inch</p>
    <footer>
      <p id="em">1em is 100% the size of the parent element</p>
      <p id="rem">1rem is 100% the size of the root element</p>
    </footer>
  </body>
</html>
```

style.css

```css
#pixel {
  font-size: 20px;
}
#point {
  font-size: 20pt;
}
#em {
  font-size: 1em;
}
#rem {
  font-size: 1rem;
}
footer {
  font-size: 12pt;
}
html {
  font-size: xx-large;
}
```



Static

1 Pixel is 1/96 of an Inch

1 Point is 1/72 of an Inch

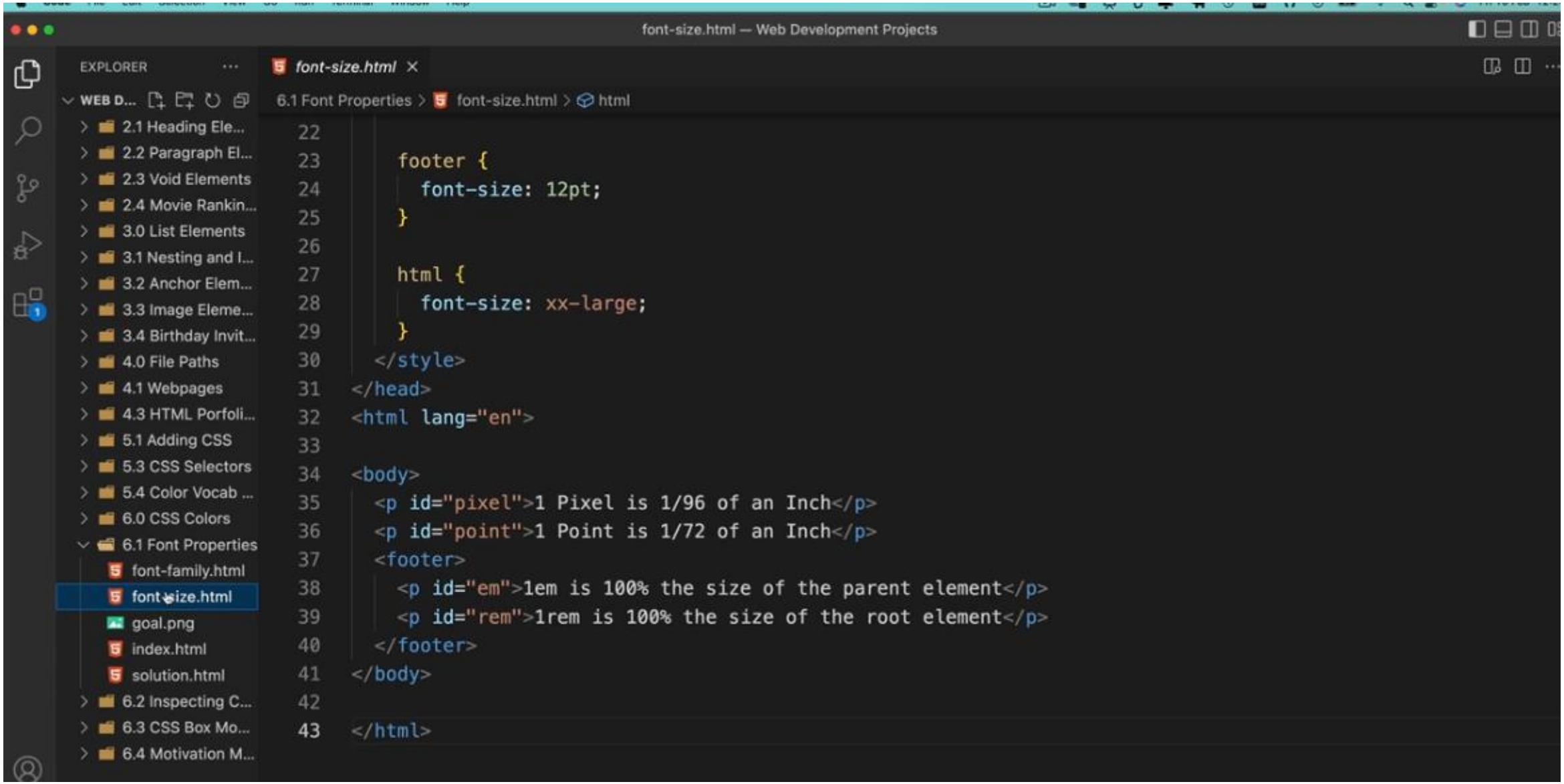
Relative

1em is 100% the size of the parent element

1rem is 100% the size of the root element



6.1 Font Properties.zip


font-size.html — Web Development Projects
EXPLORER
font-size.html
6.1 Font Properties > font-size.html > html
2.1 Heading Ele...
2.2 Paragraph El...
2.3 Void Elements
2.4 Movie Rankin...
3.0 List Elements
3.1 Nesting and I...
3.2 Anchor Elem...
3.3 Image Eleme...
3.4 Birthday Invit...
4.0 File Paths
4.1 Webpages
4.3 HTML Porfoli...
5.1 Adding CSS
5.3 CSS Selectors
5.4 Color Vocab ...
6.0 CSS Colors
6.1 Font Properties
font-family.html
font-size.html
goal.png
index.html
solution.html
6.2 Inspecting C...
6.3 CSS Box Mo...
6.4 Motivation M...
22
23 footer {
24 font-size: 12pt;
25 }
26
27 html {
28 font-size: xx-large;
29 }
30 </style>
31 </head>
32 <html lang="en">
33
34 <body>
35 1 Pixel is 1/96 of an Inch
36 1 Point is 1/72 of an Inch
37 <footer>
38 1em is 100% the size of the parent element
39 1rem is 100% the size of the root element
40 </footer>
41 </body>
42
43 </html>

Font Weight

```
h1 {
  font-weight: bold
}
```

Font Weight

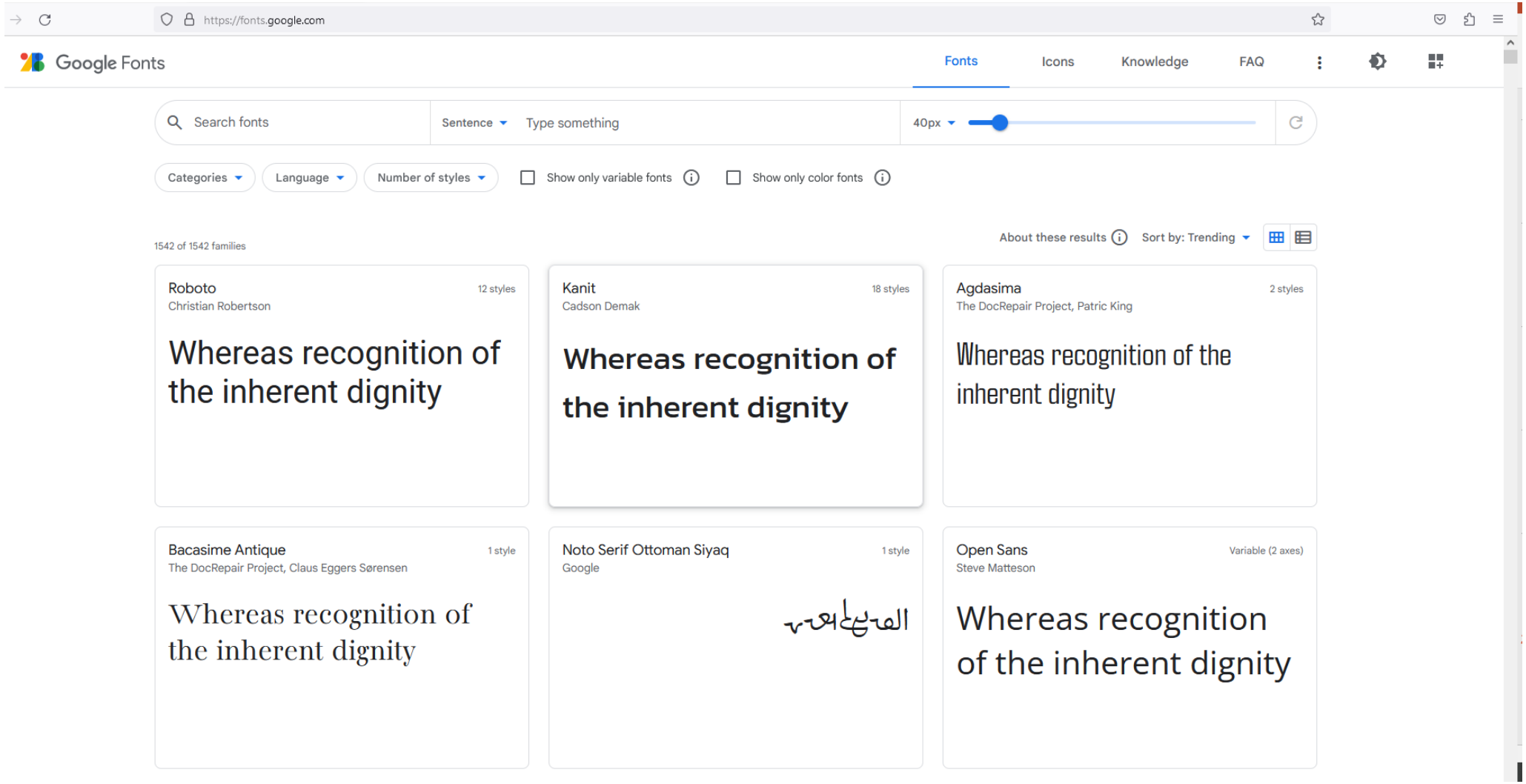
normal bold	lighter bolder	number
Keywords	Relative to Parent	100-900

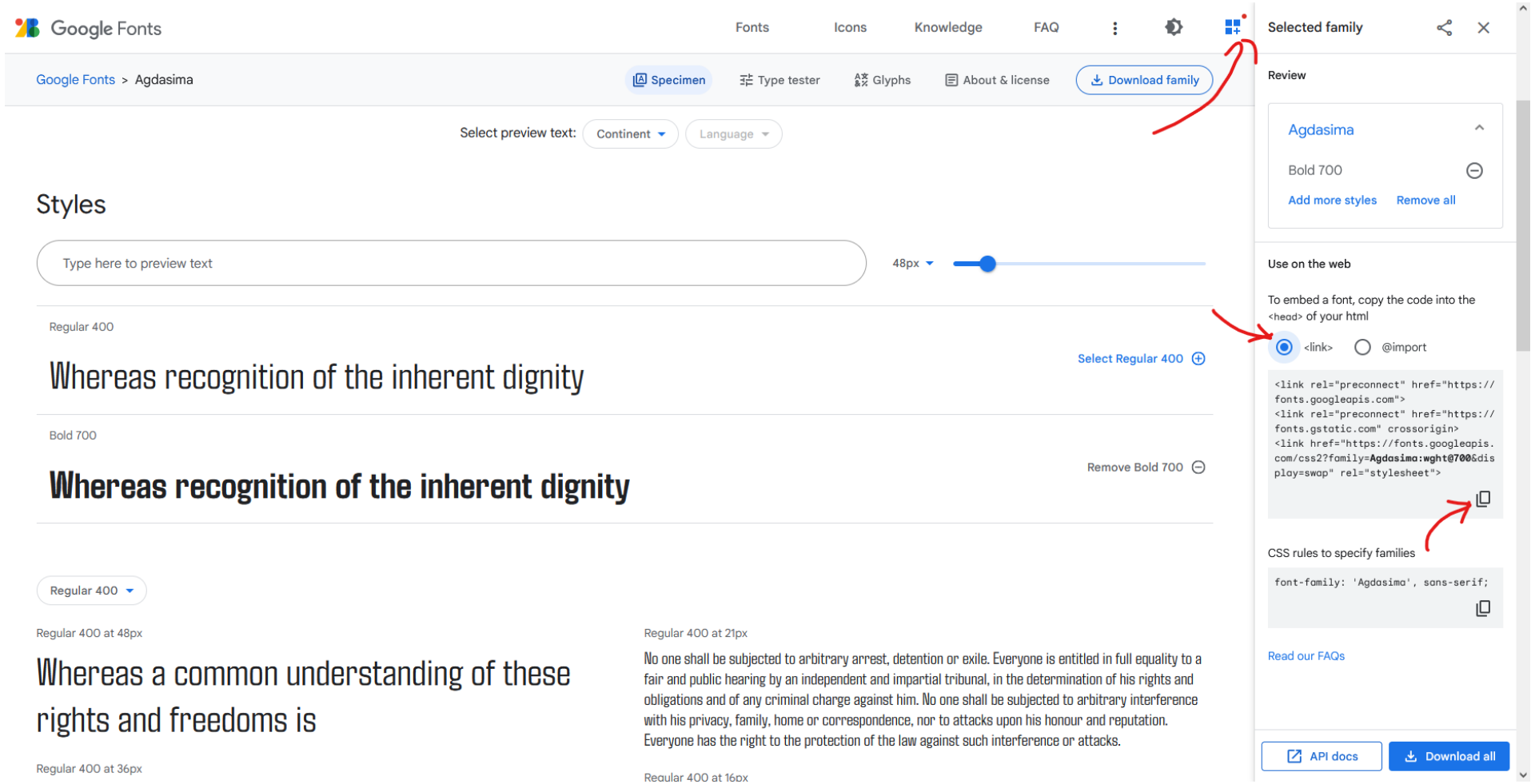
Font Family

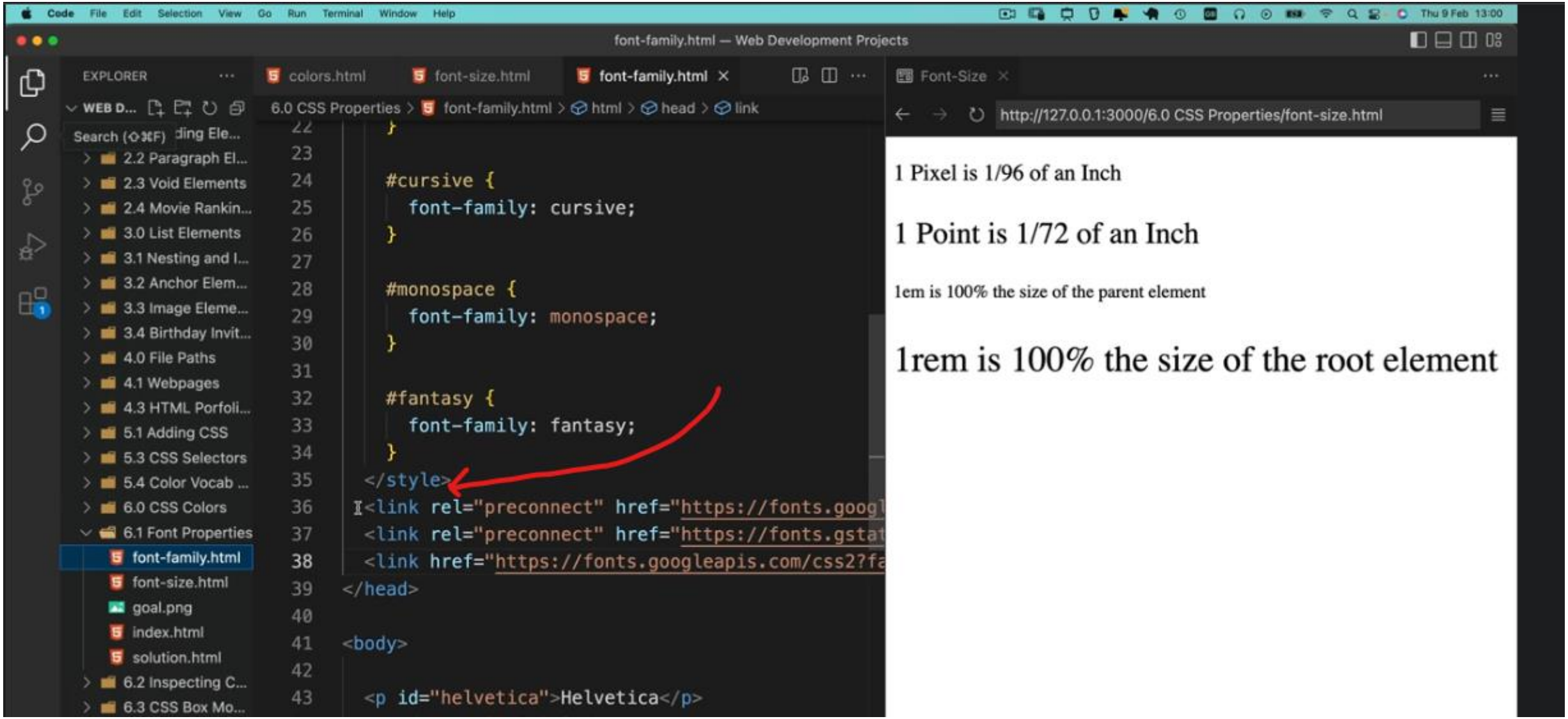
```
h1 {
  font-family: Helvetica, sans-serif
}

h2 {
  font-family: “Times New Roman”, serif
}
```

fonts.google.com



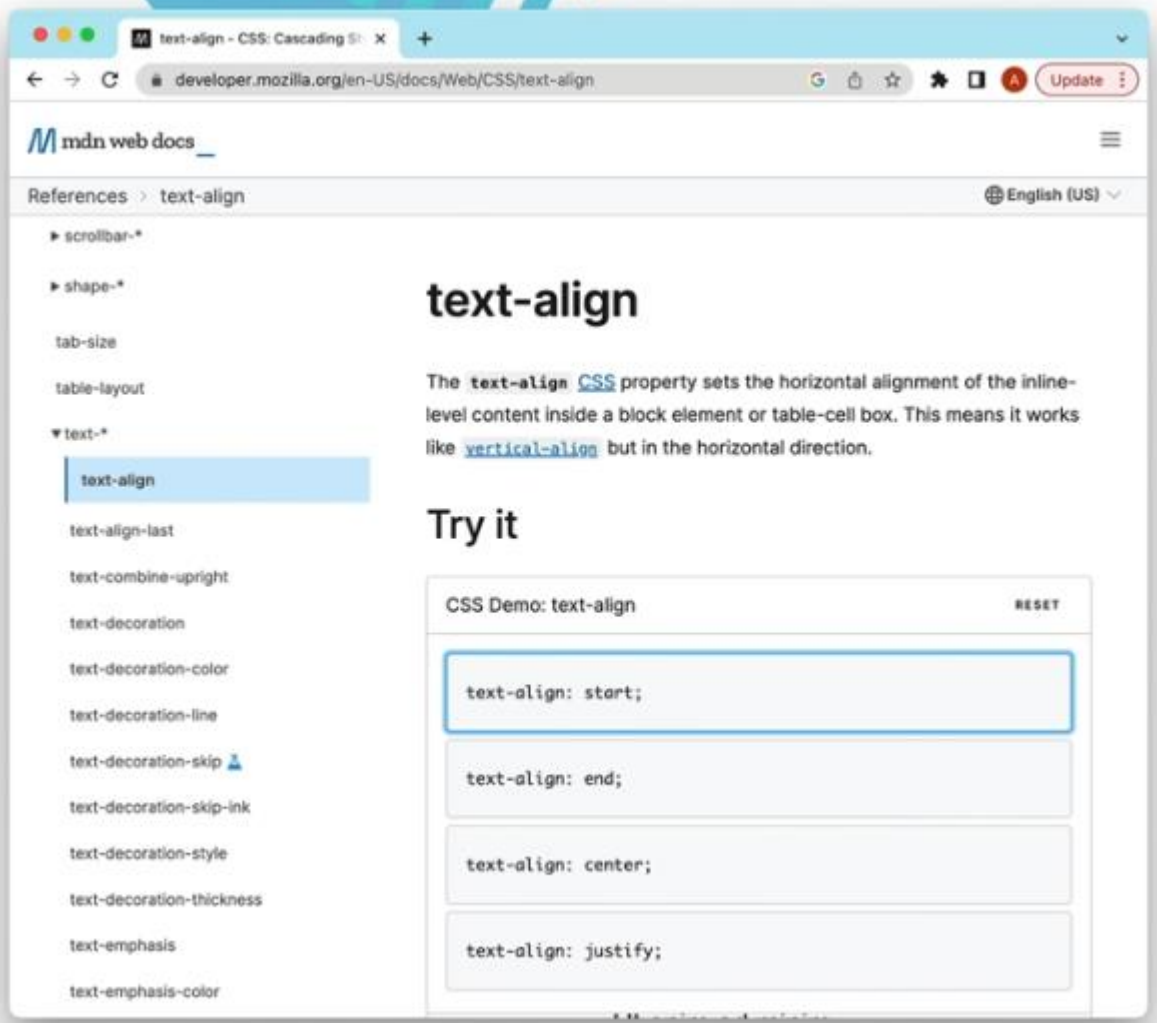

Google Fonts
Fonts
Icons
Knowledge
FAQ
Google Fonts > Agdasima
Specimen
Type tester
Glyphs
About & license
Download family
Select preview text:
Continent
Language
Styles
Type here to preview text
48px
Regular 400
Whereas recognition of the inherent dignity
Select Regular 400
Bold 700
Whereas recognition of the inherent dignity
Remove Bold 700
Regular 400
Regular 400 at 48px
Whereas a common understanding of these rights and freedoms is
Regular 400 at 36px
Regular 400 at 21px
No one shall be subjected to arbitrary arrest, detention or exile. Everyone is entitled in full equality to a fair and public hearing by an independent and impartial tribunal, in the determination of his rights and obligations and of any criminal charge against him. No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation. Everyone has the right to the protection of the law against such interference or attacks.
Regular 400 at 16px
Selected family
Review
Agdasima
Bold 700
Add more styles
Remove all
Use on the web
To embed a font, copy the code into the <head> of your html
<link>
@import
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Agdasima:wght@700&display=swap" rel="stylesheet">
CSS rules to specify families
font-family: 'Agdasima', sans-serif;
Read our FAQs
API docs
Download all


font-family.html — Web Development Projects
EXPLORER
colors.html
font-size.html
font-family.html
6.0 CSS Properties > font-family.html > html > head > link
#cursive {
font-family: cursive;
}
#monospace {
font-family: monospace;
}
#fantasy {
font-family: fantasy;
}
</style>
<link rel="preconnect" href="https://fonts.googl
<link rel="preconnect" href="https://fonts.gstat
<link href="https://fonts.googleapis.com/css2?fa
</head>
<body>
Helvetica
6.1 Font Properties
font-family.html
font-size.html
goal.png
index.html
solution.html
Font-Size
http://127.0.0.1:3000/6.0 CSS Properties/font-size.html
1 Pixel is 1/96 of an Inch
1 Point is 1/72 of an Inch
1em is 100% the size of the parent element
1rem is 100% the size of the root element

Text Align

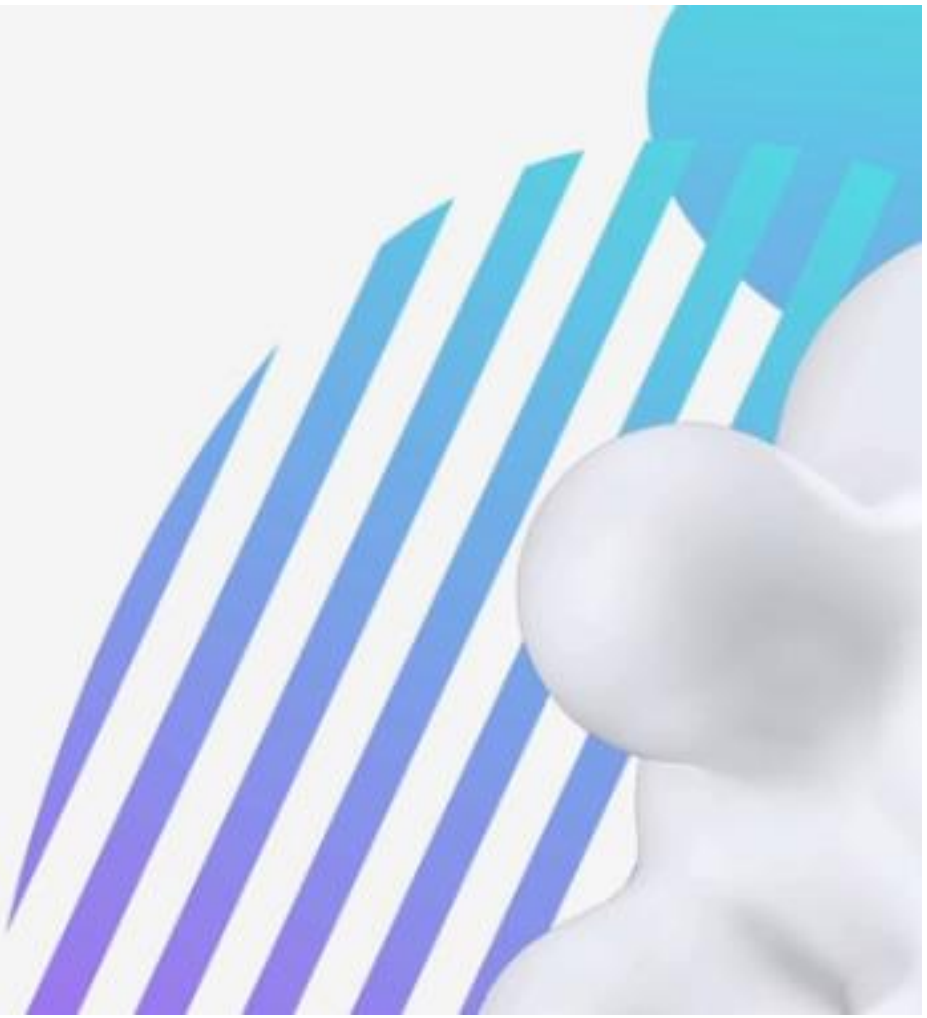
```
h1 {
  text-align: center
}
```

Left
right

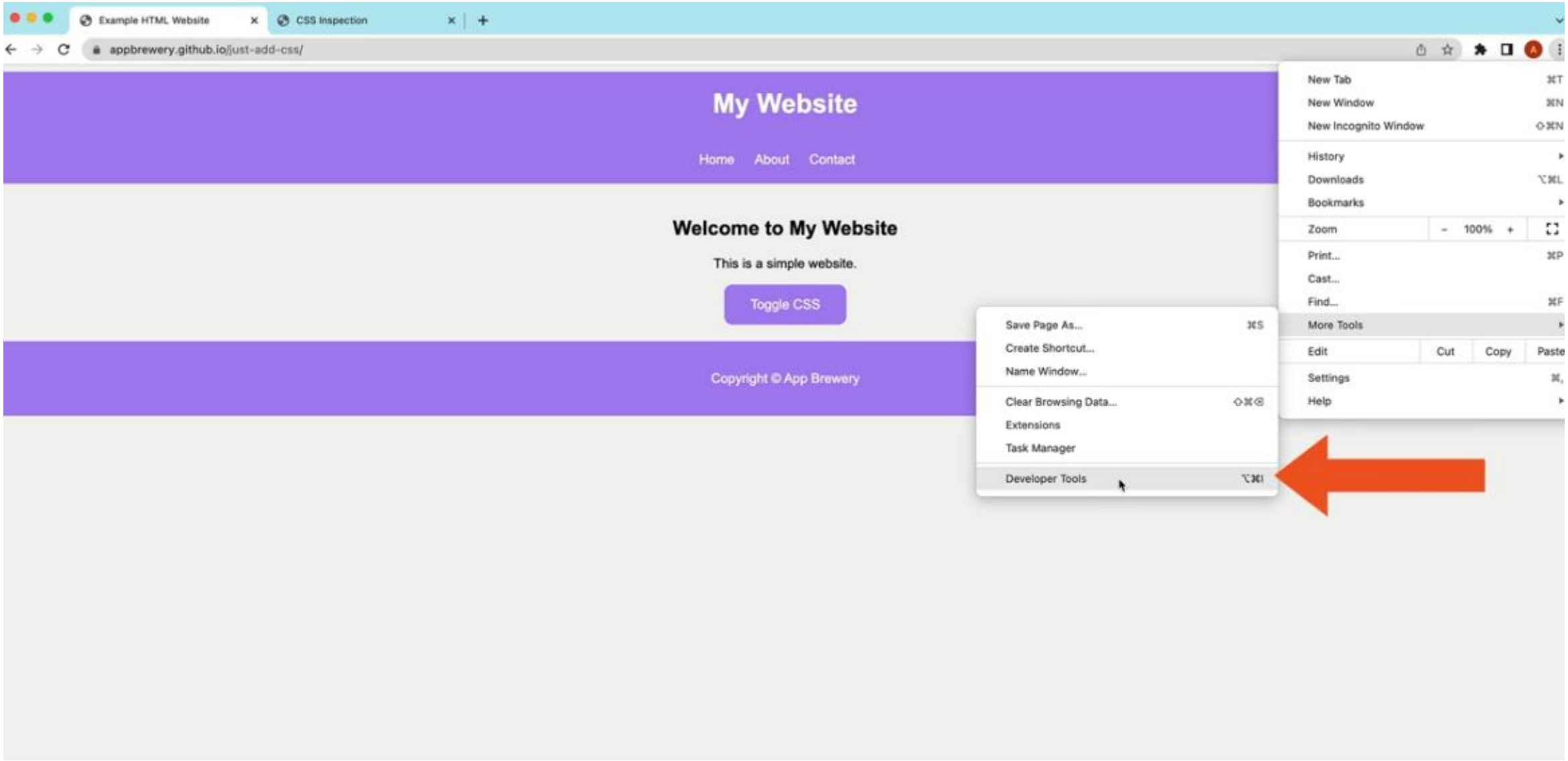


CSS Inspection

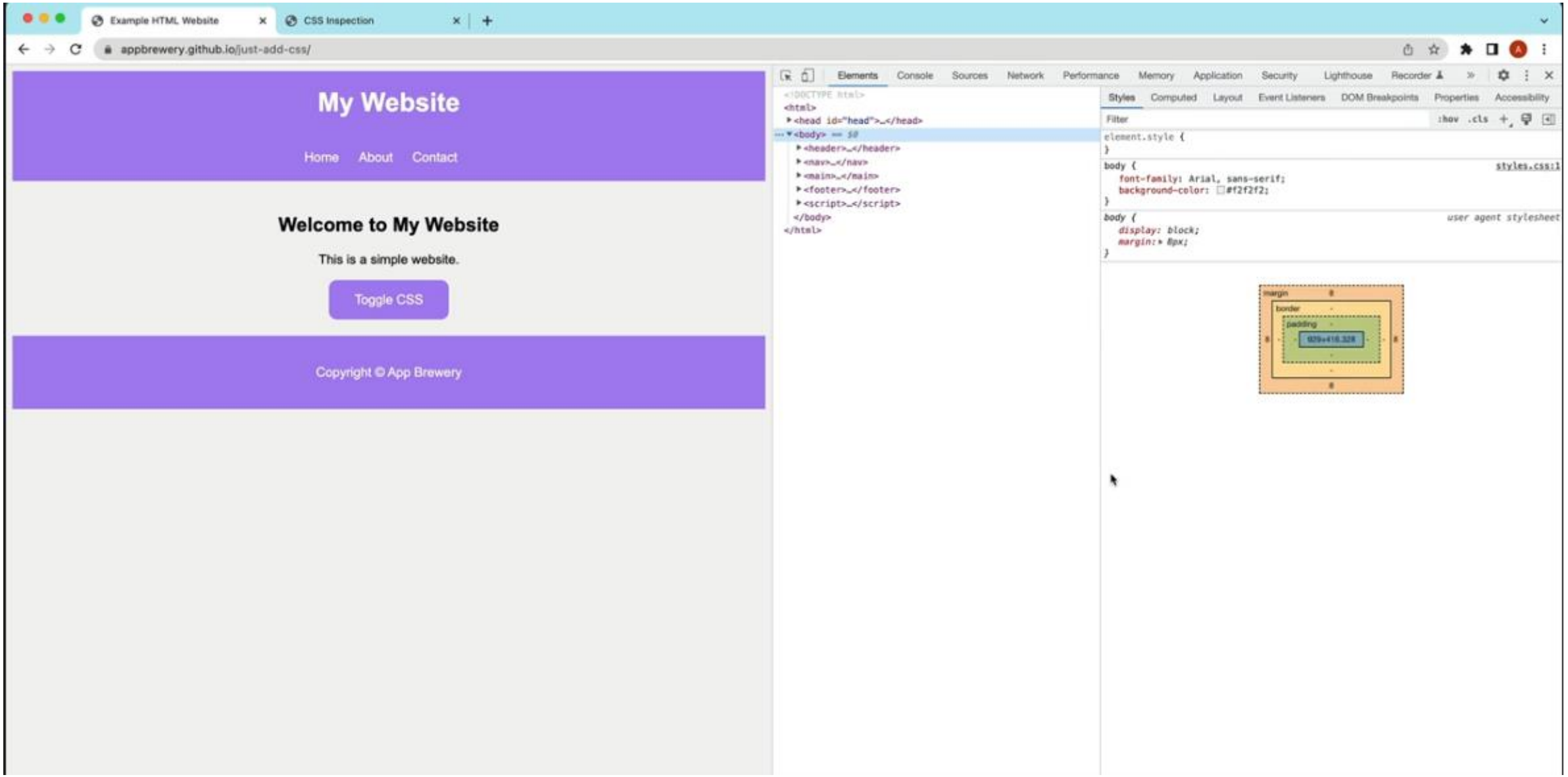
Using the CSS Overview



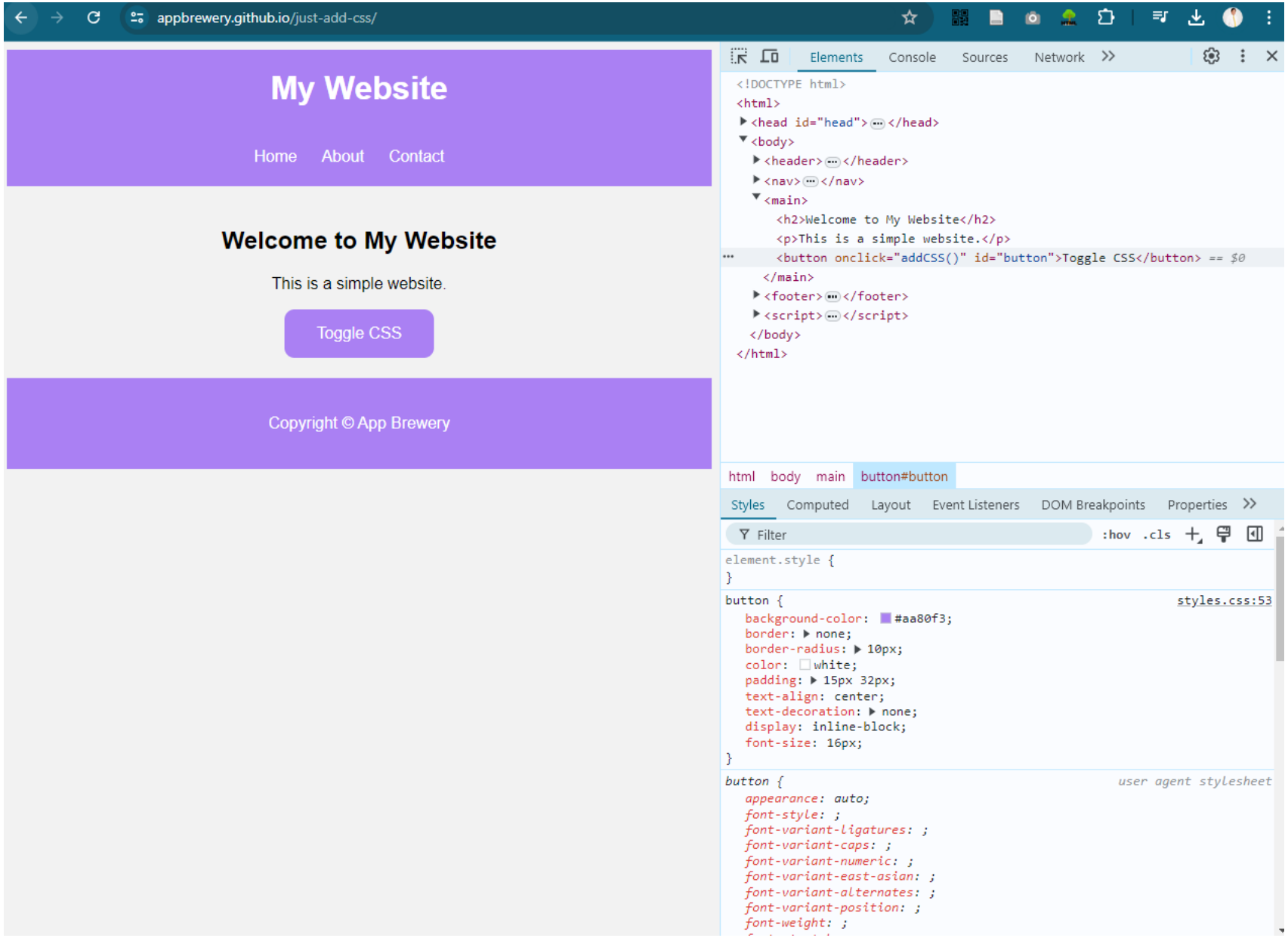
Appbrewery.github.io/just-add-css

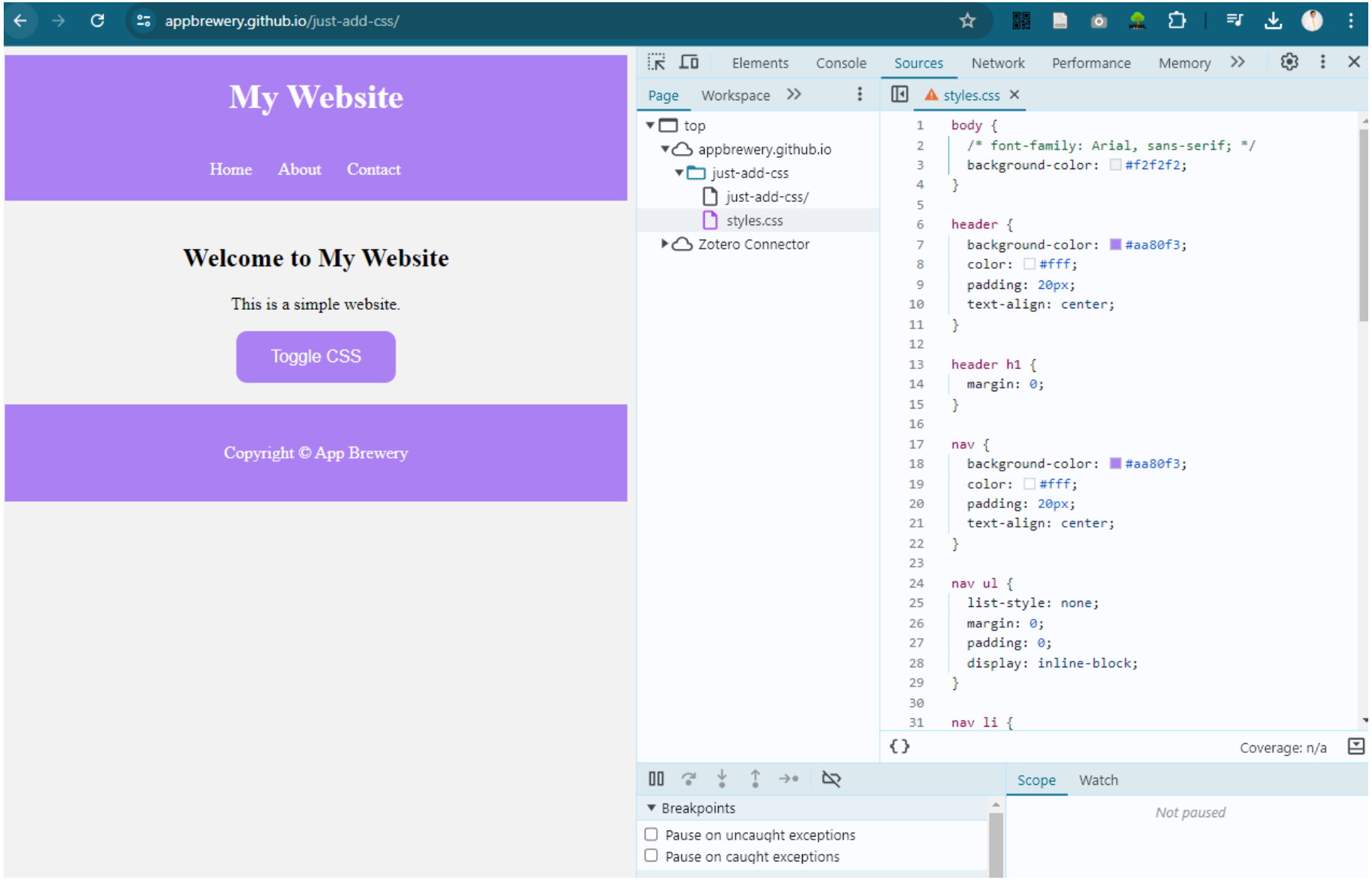



Esc
F1
F2
F3
F4
F5
F6
F7
F8
F9
F10
F11
F12
Caps Lock
Ctrl
Alt
Alt Gr
Ctrl

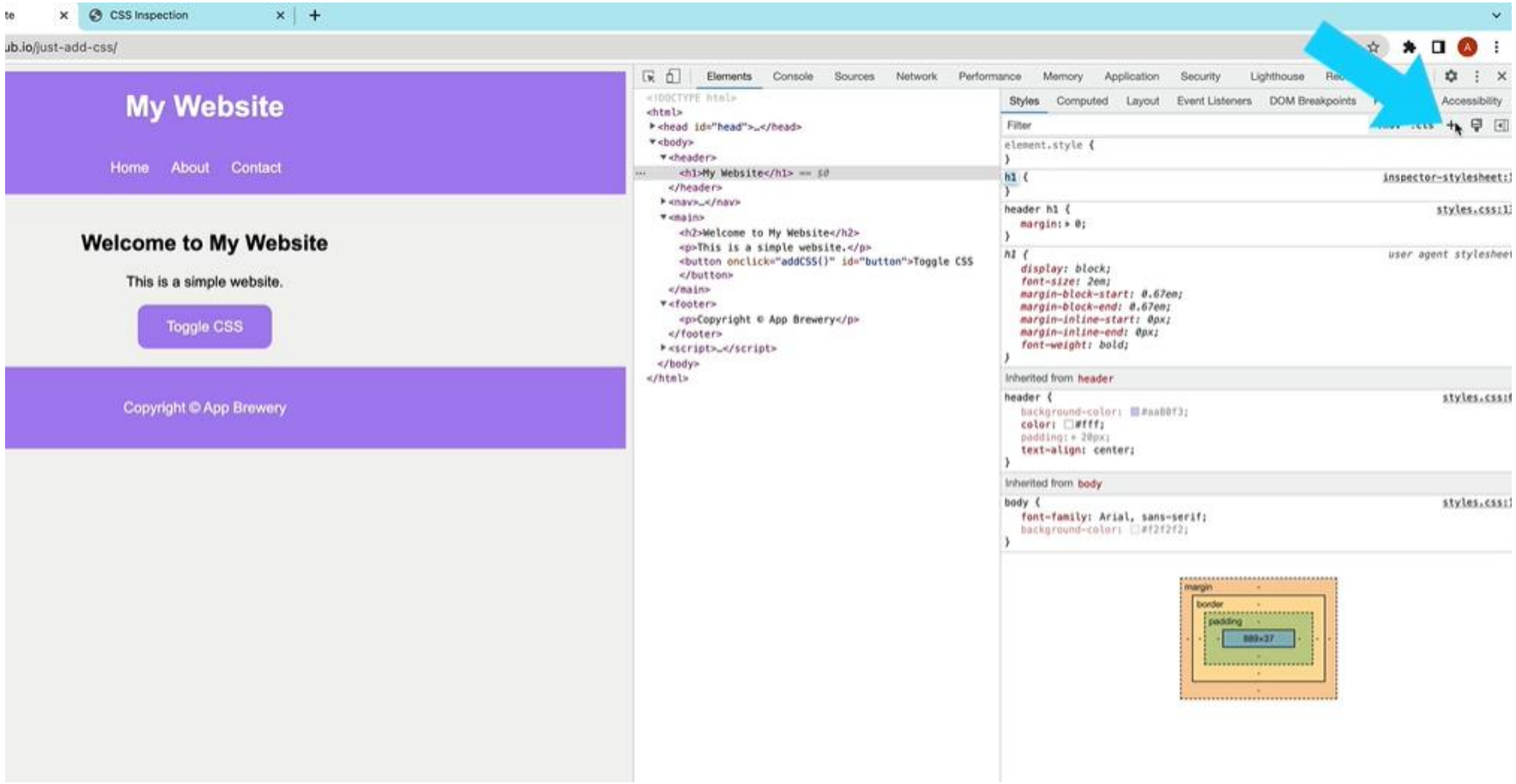


https://appbrewery.github.io/css-inspection/

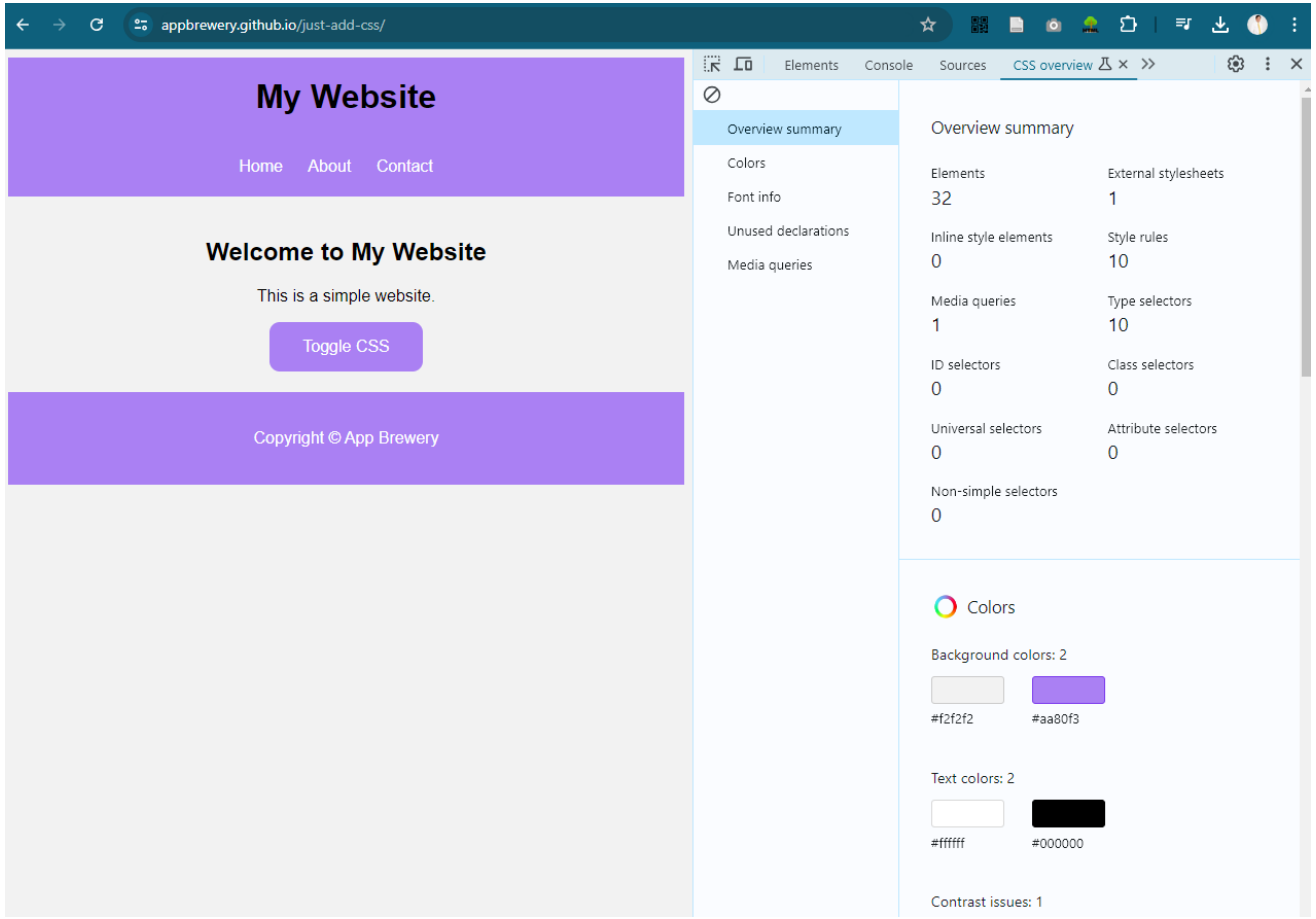

appbrewery.github.io/just-add-css/
My Website
Home About Contact
Welcome to My Website
This is a simple website.
Toggle CSS
Copyright © App Brewery
Elements Console Sources Network
<!DOCTYPE html>
<html>
<head id="head">...</head>
<body>
<header>...</header>
<nav>...</nav>
<main>
<h2>Welcome to My Website</h2>
This is a simple website.
<button onclick="addCSS()" id="button">Toggle CSS</button> == $0
</main>
<footer>...</footer>
<script>...</script>
</body>
</html>
html body main button#button
Styles Computed Layout Event Listeners DOM Breakpoints Properties
Filter
:hov .cls
element.style {
}
button { styles.css:53
background-color: #aa80f3;
border: none;
border-radius: 10px;
color: white;
padding: 15px 32px;
text-align: center;
text-decoration: none;
display: inline-block;
font-size: 16px;
}
button { user agent stylesheet
appearance: auto;
font-style: ;
font-variant-ligatures: ;
font-variant-caps: ;
font-variant-numeric: ;
font-variant-east-asian: ;
font-variant-alternates: ;
font-variant-position: ;
font-weight: ;


appbrewery.github.io/just-add-css/
My Website
Home
About
Contact
Welcome to My Website
This is a simple website.
Toggle CSS
Copyright © App Brewery
Elements
Console
Sources
Network
Performance
Memory
Page
Workspace
styles.css
top
appbrewery.github.io
just-add-css
just-add-css/
styles.css
Zotero Connector
1 body {
2 /* font-family: Arial, sans-serif; */
3 background-color: #f2f2f2;
4 }
5
6 header {
7 background-color: #aa80f3;
8 color: #fff;
9 padding: 20px;
10 text-align: center;
11 }
12
13 header h1 {
14 margin: 0;
15 }
16
17 nav {
18 background-color: #aa80f3;
19 color: #fff;
20 padding: 20px;
21 text-align: center;
22 }
23
24 nav ul {
25 list-style: none;
26 margin: 0;
27 padding: 0;
28 display: inline-block;
29 }
30
31 nav li {
{}
Coverage: n/a
Scope
Watch
Not paused
Breakpoints
Pause on uncaught exceptions
Pause on caught exceptions

Add CSS


CSS Inspection
My Website
Home About Contact
Welcome to My Website
This is a simple website.
Toggle CSS
Copyright © App Brewery
Elements Console Sources Network Performance Memory Application Security Lighthouse
Styles Computed Layout Event Listeners DOM Breakpoints Accessibility
Filter
element.style {
}
h1 {
}
header h1 {
margin: ▸ 0;
}
h1 {
display: block;
font-size: 2em;
margin-block-start: 0.67em;
margin-block-end: 0.67em;
margin-inline-start: 0px;
margin-inline-end: 0px;
font-weight: bold;
}
Inherited from header
header {
background-color: #aa80f3;
color: #fff;
padding: ▸ 20px;
text-align: center;
}
Inherited from body
body {
font-family: Arial, sans-serif;
background-color: #f2f2f2;
}
margin
border
padding
889×37

CSS overview tool




appbrewery.github.io/css-inspection/
Inspect Me
This is an simple website to learn how to do CSS Inspection
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nam viverra euismod odio, gravida pellentesque urna varius vitae. Suspendisse potenti. Morbi commodo, ipsum in blandit hendrerit, velit enim ornare nisi, eget consequat libero
eget ornare commodo, quam libero pretium libero, euismod blandit mauris nunc in magna.
Q.1 What is the named color of the body?
Submit
Q.2 What is the font size of the h1?
Submit
Q.3 What is the font weight of the h2?
Submit
Q.4 What is the font family of the p?
Submit

The Box Model

Margin, Padding and Border





```
.Cat{
    width: 100%;
    height: 100%;
}
```

https://www.ilovebacon.com

I Love Bacon

bacon, bacon, bacon, bacon, bacon, bacon
bacon, bacon, bacon, bacon, bacon, bacon
bacon, bacon, bacon, bacon, bacon, bacon

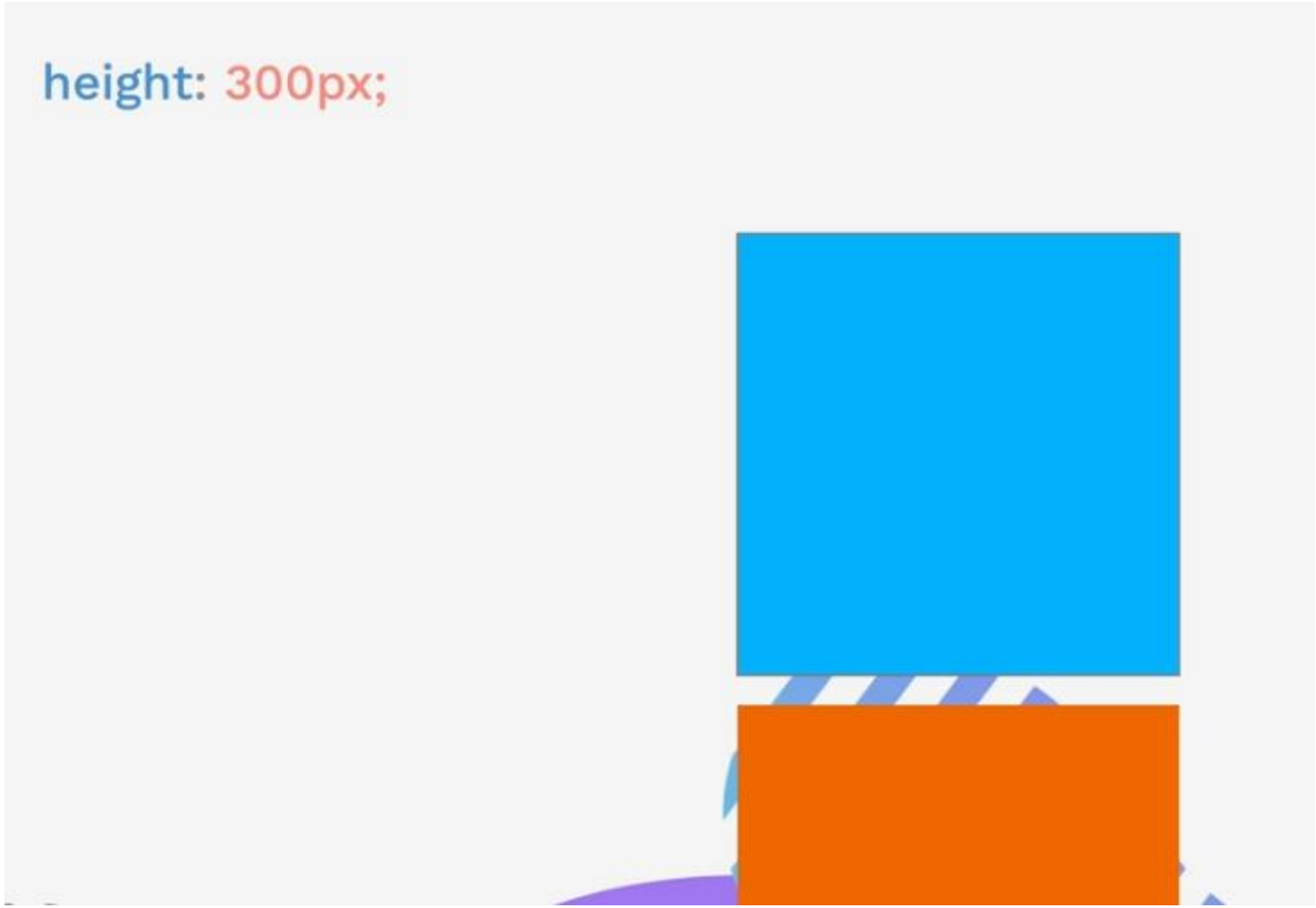


https://www.ilovebacon.com

I Love Bacon

bacon, bacon, bacon, bacon, bacon, bacon
bacon, bacon, bacon, bacon, bacon, bacon
bacon, bacon, bacon, bacon, bacon, bacon



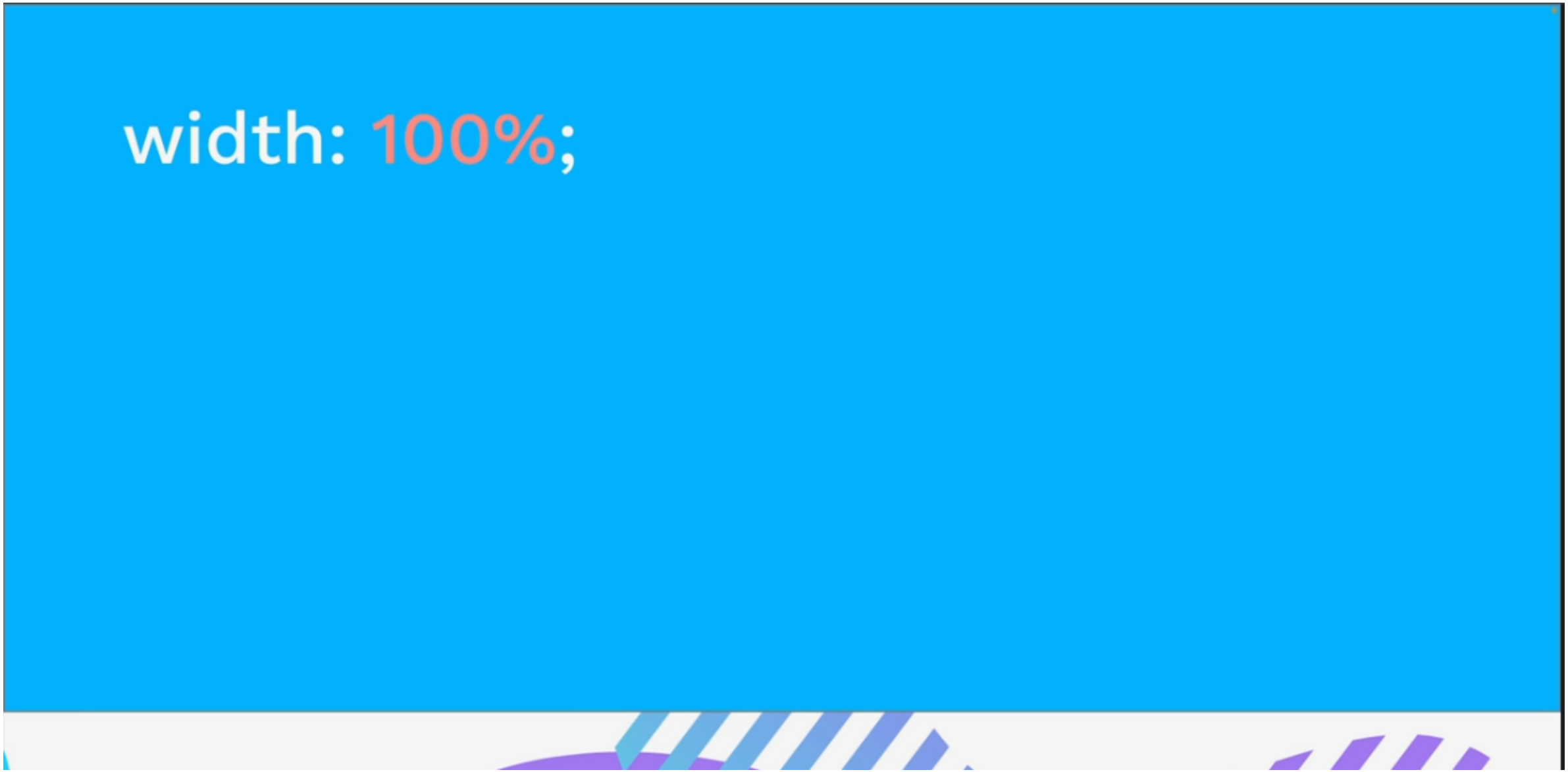

height: 300px;


height: 600px;

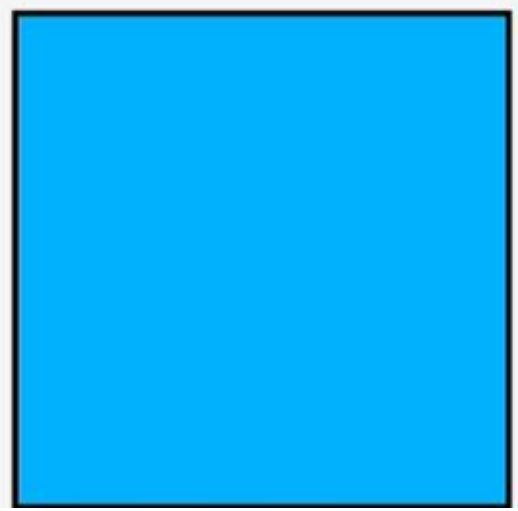
height: 600px;
width: 600px;



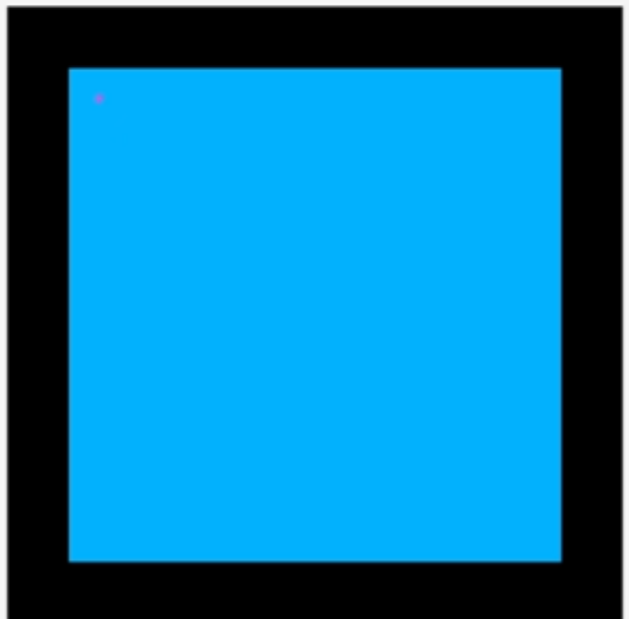
The


width: 100%;

border: 10px solid black;



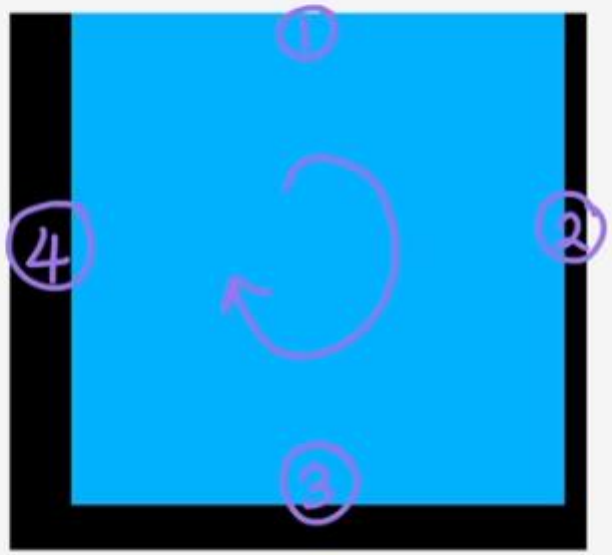
border: 30px solid black;

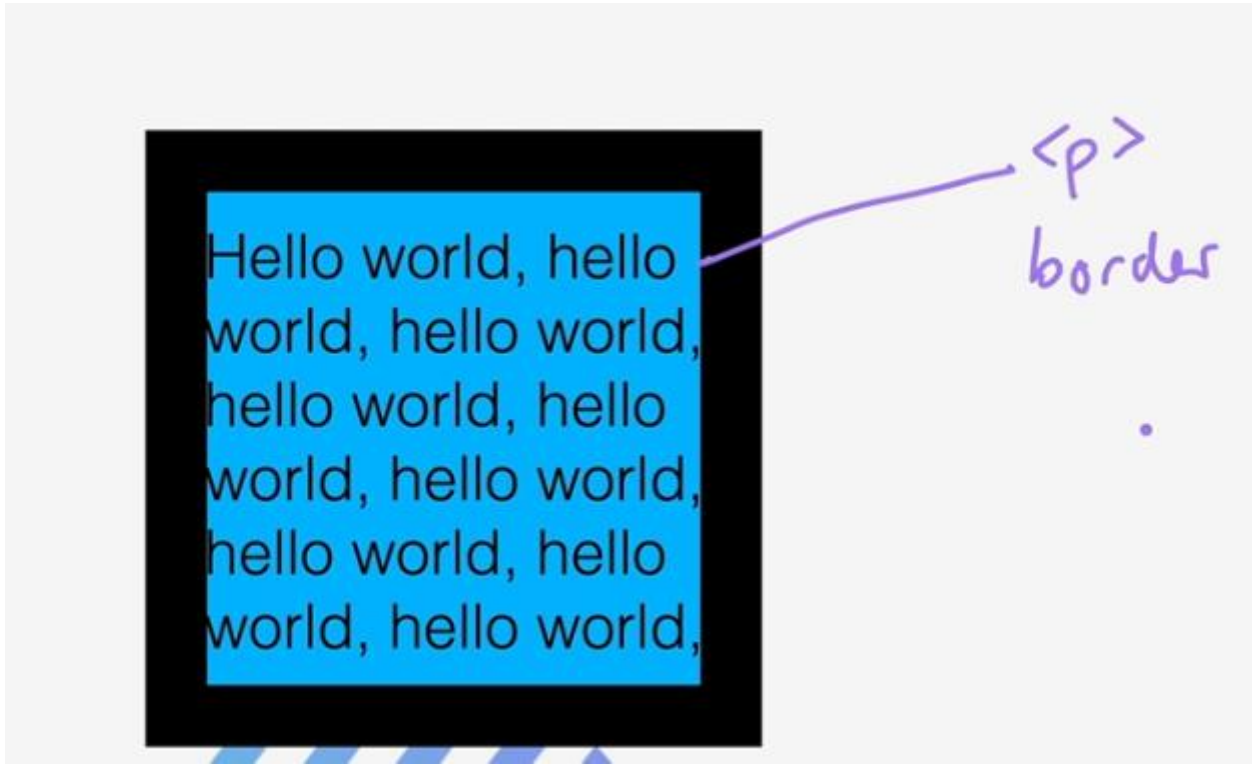


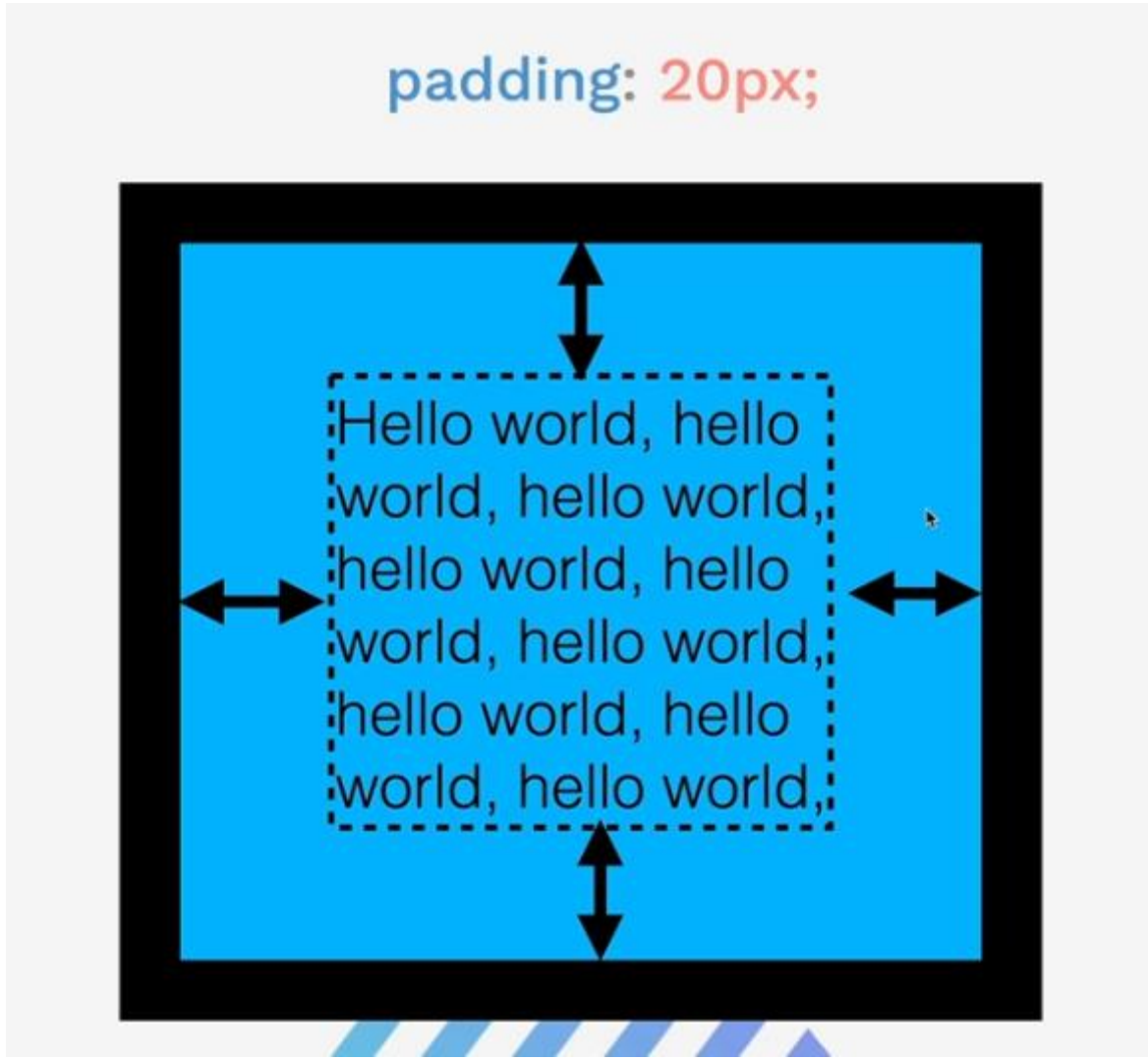
- border: 30px solid black;
  border-top: 0px;

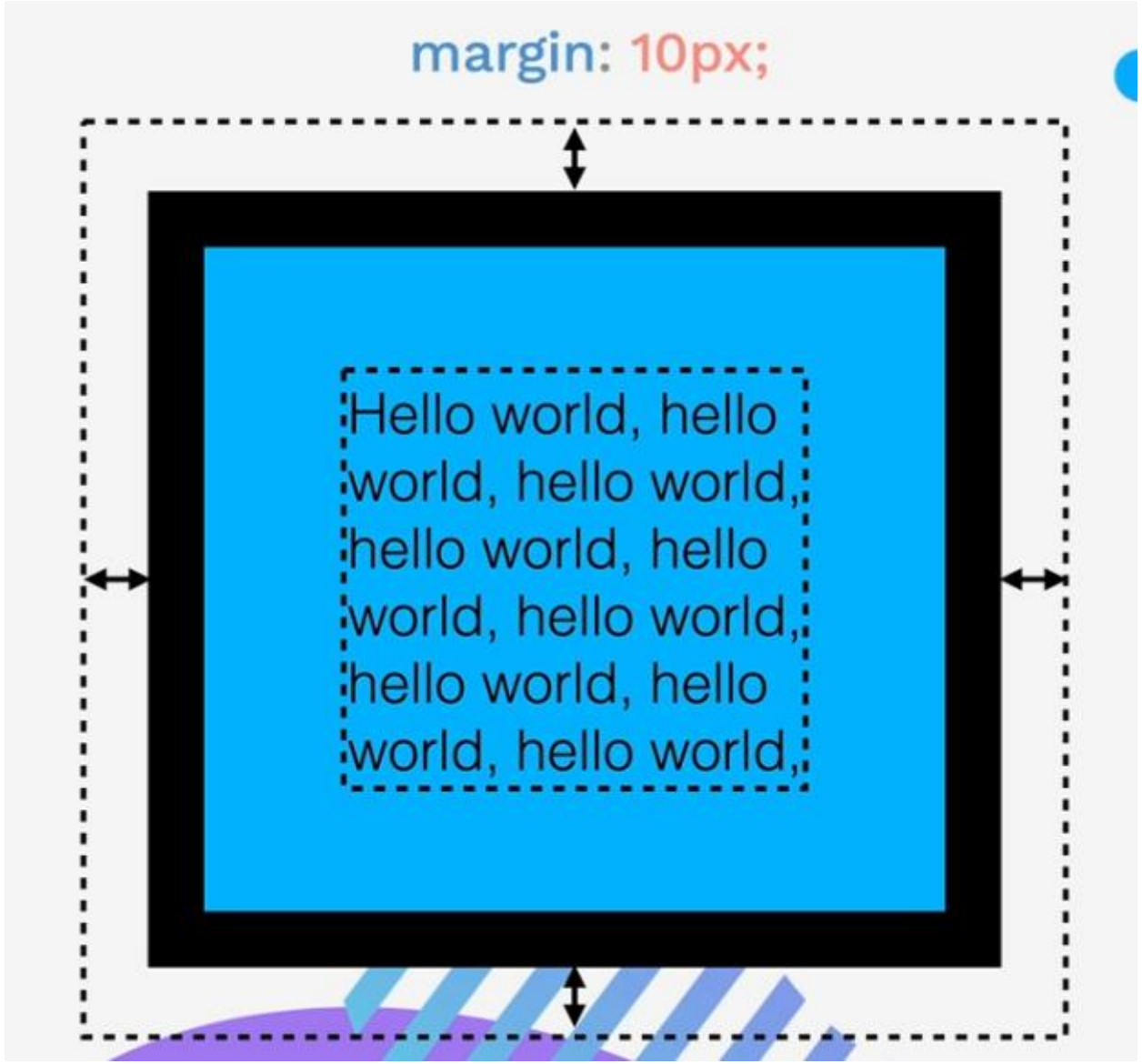


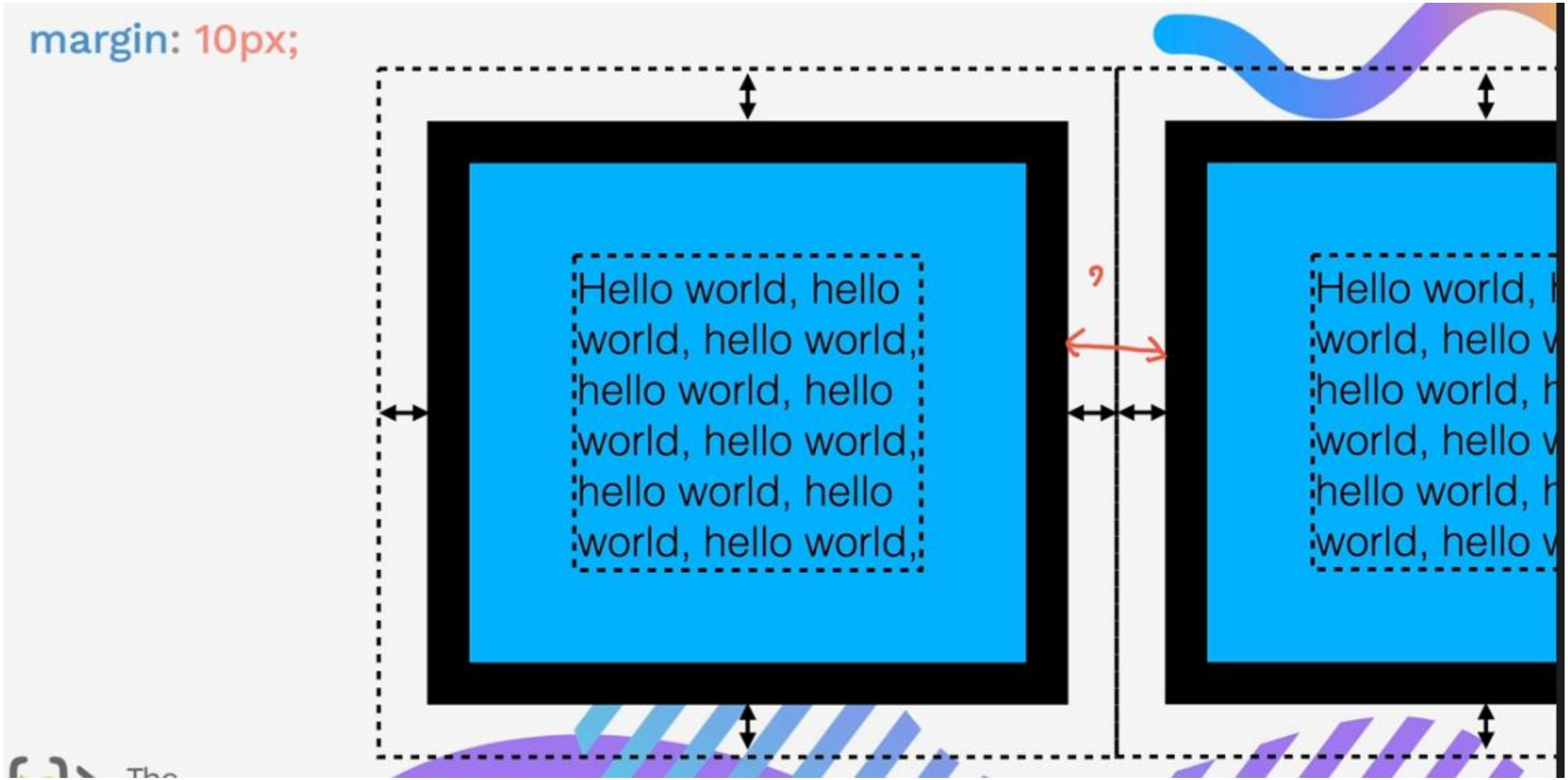
```
border: 30px solid black;
border-width: 0px 10px 20px 30px;
```

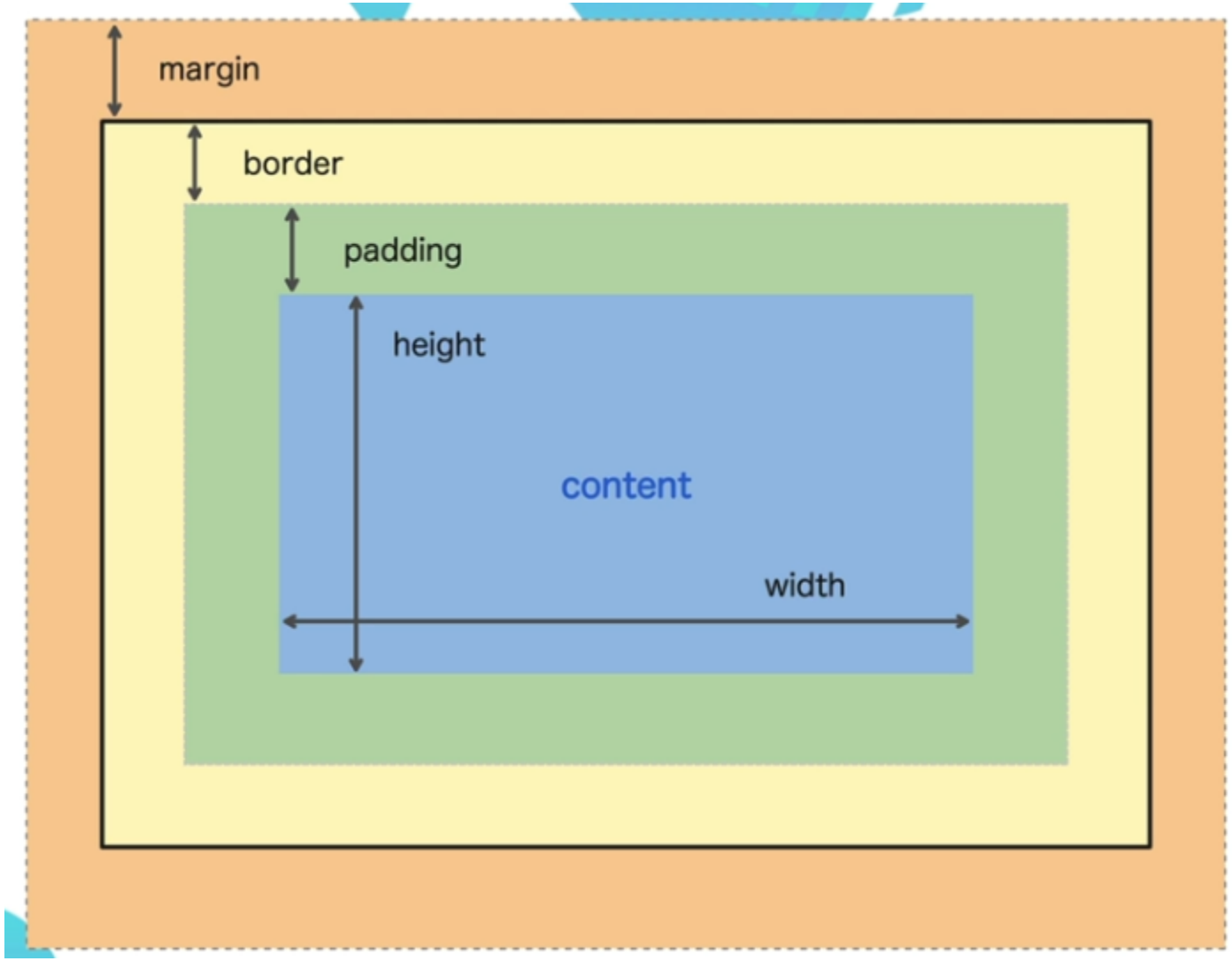


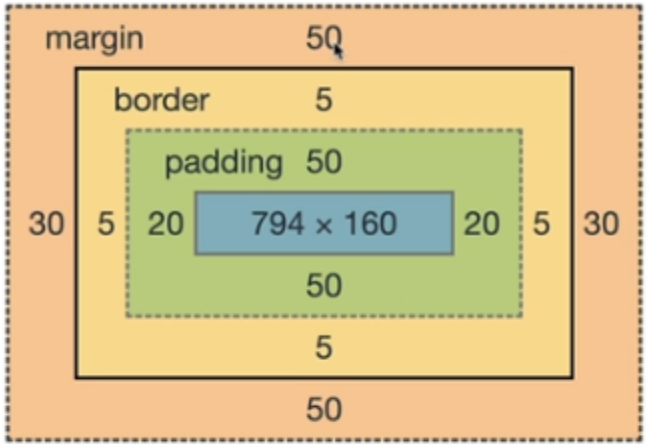

Hello world, hello world, hello world, hello world, hello world, hello world, hello world, hello world,
<p>
border


padding: 20px;
Hello world, hello world, hello world, hello world, hello world, hello world, hello world, hello world,


margin: 10px;
Hello world, hello world, hello world, hello world, hello world, hello world, hello world, hello world,


margin: 10px;
Hello world, hello world, hello world, hello world, hello world, hello world, hello world, hello world,
?
Hello world, h
world, hello w
hello world, h
world, hello w
hello world, h
world, hello w


margin
border
padding
height
content
width


margin
50
border
5
padding
50
30
5
20
794 × 160
20
5
30
50
5
50

CSS Box Model

Pesticide for Chrome - Chrome

appbrewery.github.io/box-model/

Hello World



Good Night World



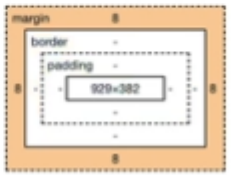
Elements Console Sources Network Performance Memory Application Security Lighthouse Recorder

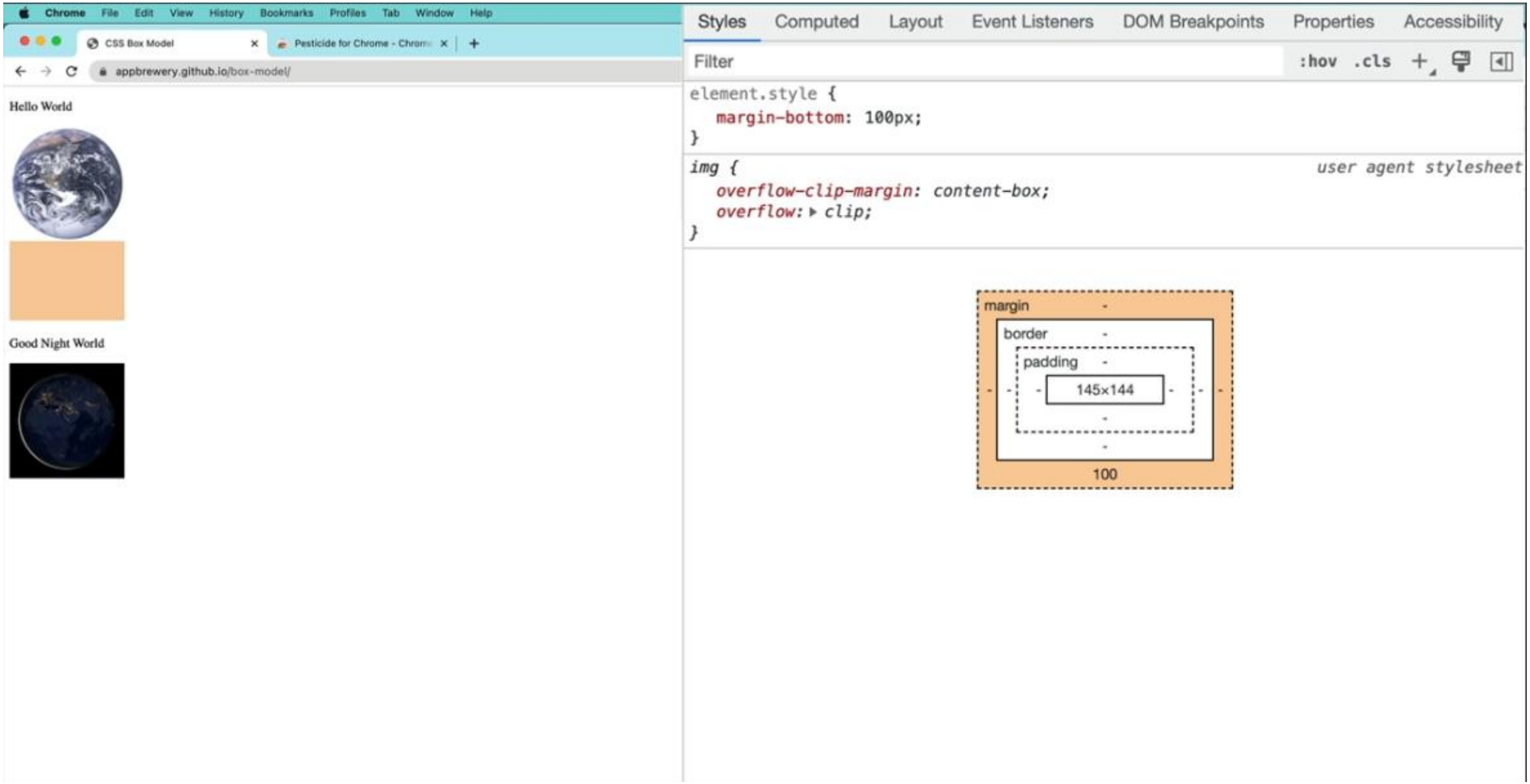
```
<!DOCTYPE html>
<html lang="en">
▶ <head>…</head>
… ▼ <body> == $0
  ▶ <div>…</div>
  ▶ <div>…</div>
  </body>
</html>
```

Styles Computed Layout Event Listeners DOM Breakpoints Properties Accessibility

Filter :hov .cls

```
element.style {
}
body {                                  user agent stylesheet
    display: block;
    margin: ▸ 8px;
}
```




Chrome File Edit View History Bookmarks Profiles Tab Window Help
CSS Box Model
Pesticide for Chrome - Chrom
appbrewery.github.io/box-model/
Hello World
Good Night World
Styles
Computed
Layout
Event Listeners
DOM Breakpoints
Properties
Accessibility
Filter
:hov .cls
element.style {
margin-bottom: 100px;
}
img { user agent stylesheet
overflow-clip-margin: content-box;
overflow: clip;
}
margin
border
padding
145×144
100

https://www.ilovebacon.com

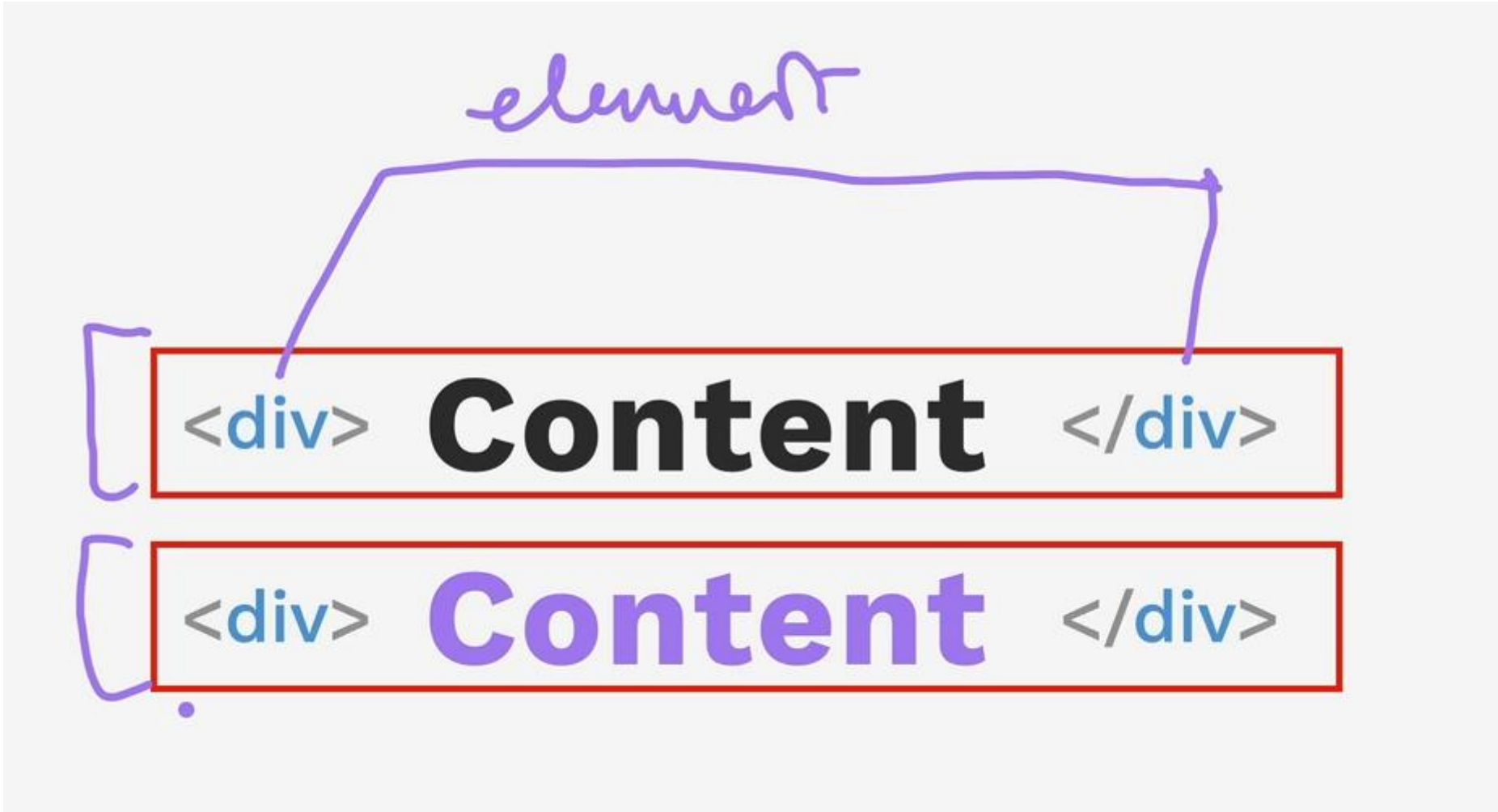
I Love Bacon

bacon, bacon, bacon, bacon, bacon, bacon
bacon, bacon, bacon, bacon, bacon, bacon
bacon, bacon, bacon, bacon, bacon, bacon



This is bacon

Content Division Element


element
 Content
 Content

```html
<body>
  <div>
    <p>Hello World</p>
    <img src="./world.png"/>
  </div>
  <div>
    <p>Good Night World</p>
    <img src="./night-world.png"/>
  </div>
</body>
```

Hello World



Good Night World



Pesticide Chrome Extension



chrome.google.com/webstore/detail/pesticide-for-chrome/bakpbgckdnepkmkeaiomhmfcnejndkbi

chrome web store

Home > Extensions > Pesticide for Chrome

Pesticide for Chrome

★★★★★ 35 | Developer Tools | 100,000+ users

Add to Chrome

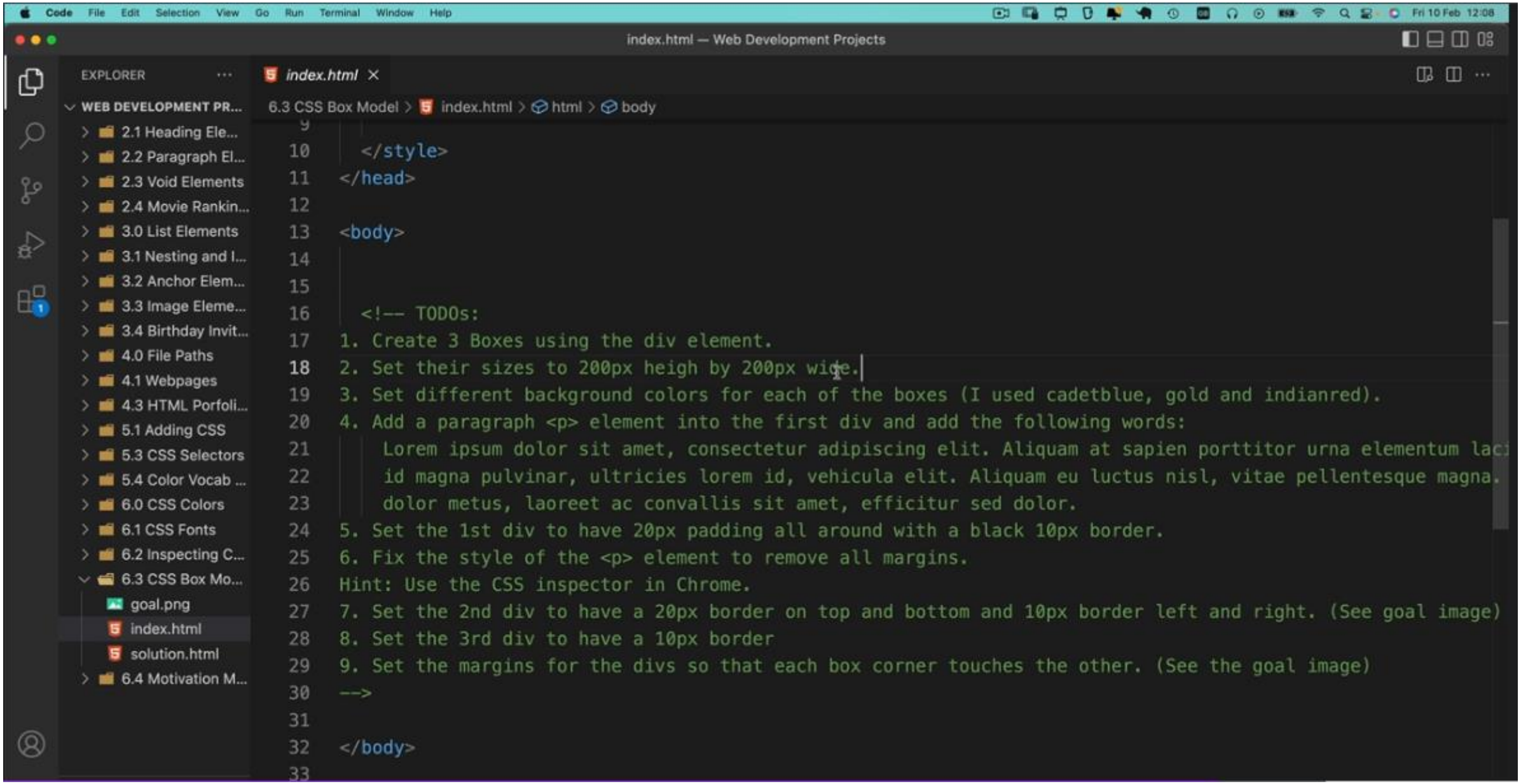
Overview | Privacy practices | Reviews | Related



Try an Exercise



6.3 CSS
Box Model.zip

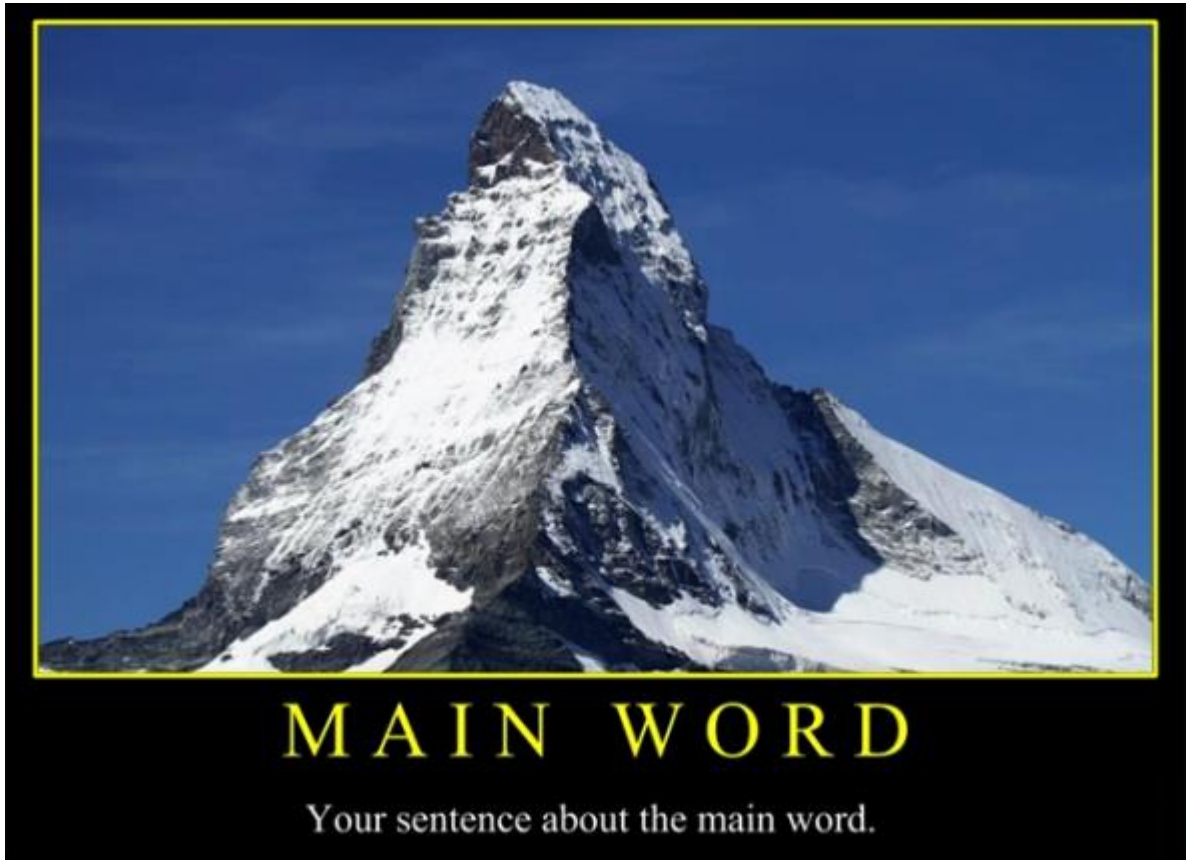

Code File Edit Selection View Go Run Terminal Window Help
Fri 10 Feb 12:08
index.html — Web Development Projects
EXPLORER
WEB DEVELOPMENT PR...
2.1 Heading Ele...
2.2 Paragraph El...
2.3 Void Elements
2.4 Movie Rankin...
3.0 List Elements
3.1 Nesting and I...
3.2 Anchor Elem...
3.3 Image Eleme...
3.4 Birthday Invit...
4.0 File Paths
4.1 Webpages
4.3 HTML Porfoli...
5.1 Adding CSS
5.3 CSS Selectors
5.4 Color Vocab ...
6.0 CSS Colors
6.1 CSS Fonts
6.2 Inspecting C...
6.3 CSS Box Mo...
goal.png
index.html
solution.html
6.4 Motivation M...
index.html
6.3 CSS Box Model > index.html > html > body
10 </style>
11 </head>
12
13 <body>
14
15
16 <!-- TODOs:
17 1. Create 3 Boxes using the div element.
18 2. Set their sizes to 200px heigh by 200px wide.
19 3. Set different background colors for each of the boxes (I used cadetblue, gold and indianred).
20 4. Add a paragraph <p> element into the first div and add the following words:
21 Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam at sapien porttitor urna elementum laci
22 id magna pulvinar, ultricies lorem id, vehicula elit. Aliquam eu luctus nisl, vitae pellentesque magna.
23 dolor metus, laoreet ac convallis sit amet, efficitur sed dolor.
24 5. Set the 1st div to have 20px padding all around with a black 10px border.
25 6. Fix the style of the <p> element to remove all margins.
26 Hint: Use the CSS inspector in Chrome.
27 7. Set the 2nd div to have a 20px border on top and bottom and 10px border left and right. (See goal image)
28 8. Set the 3rd div to have a 10px border
29 9. Set the margins for the divs so that each box corner touches the other. (See the goal image)
30 -->
31
32 </body>
33

Motivational Poster Project

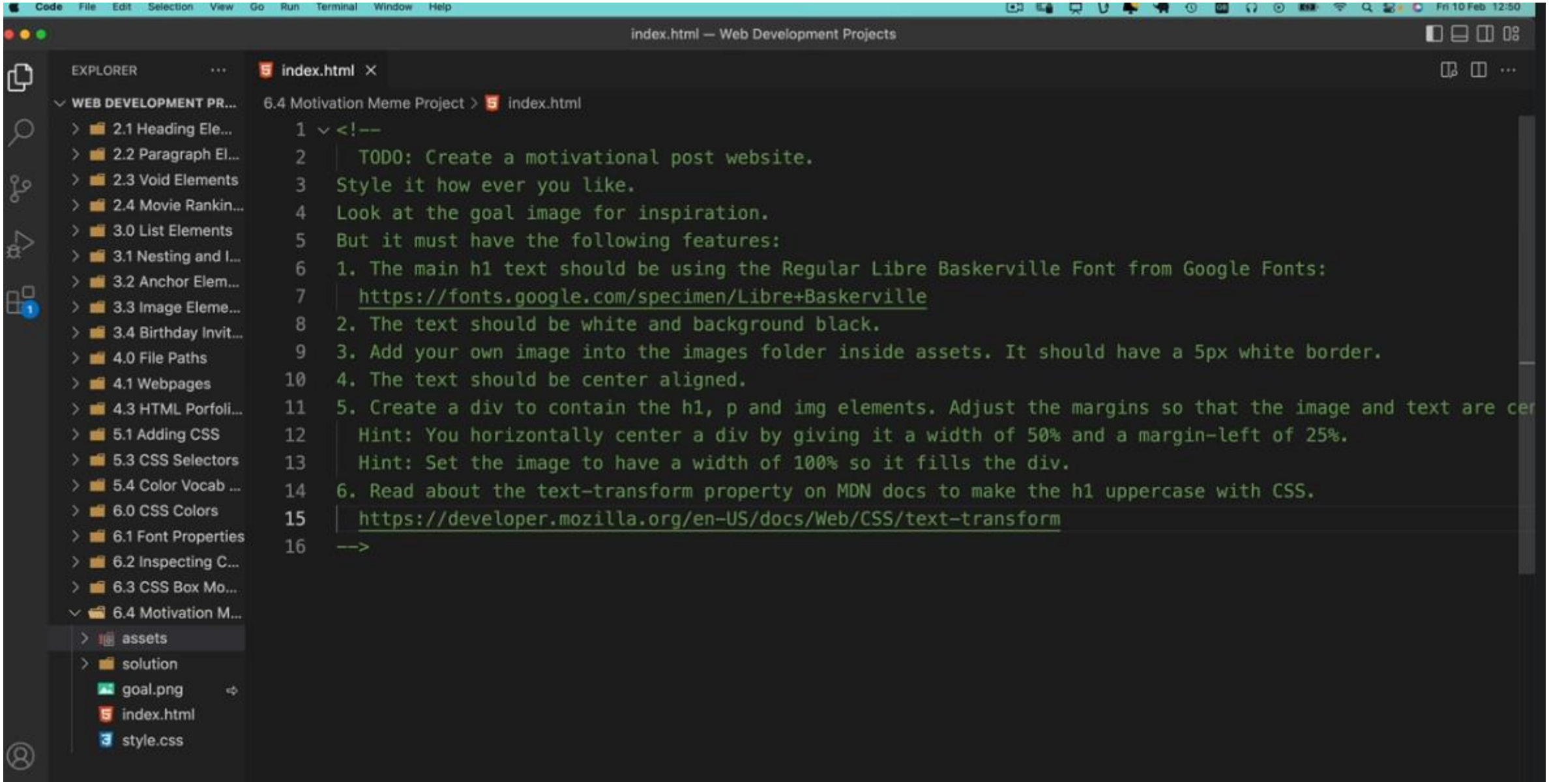
Make your own memes




POTENTIAL
We all have the tools for greatness within us.


MAIN WORD
Your sentence about the main word.


ZIP
Motivation Meme
Project.zip


Code File Edit Selection View Go Run Terminal Window Help
Fri 10 Feb 12:50
index.html — Web Development Projects
EXPLORER
index.html
WEB DEVELOPMENT PR...
2.1 Heading Ele...
2.2 Paragraph El...
2.3 Void Elements
2.4 Movie Rankin...
3.0 List Elements
3.1 Nesting and I...
3.2 Anchor Elem...
3.3 Image Eleme...
3.4 Birthday Invit...
4.0 File Paths
4.1 Webpages
4.3 HTML Porfoli...
5.1 Adding CSS
5.3 CSS Selectors
5.4 Color Vocab ...
6.0 CSS Colors
6.1 Font Properties
6.2 Inspecting C...
6.3 CSS Box Mo...
6.4 Motivation M...
assets
solution
goal.png
index.html
style.css
6.4 Motivation Meme Project > index.html
<!--
TODO: Create a motivational post website.
Style it how ever you like.
Look at the goal image for inspiration.
But it must have the following features:
1. The main h1 text should be using the Regular Libre Baskerville Font from Google Fonts:
https://fonts.google.com/specimen/Libre+Baskerville
2. The text should be white and background black.
3. Add your own image into the images folder inside assets. It should have a 5px white border.
4. The text should be center aligned.
5. Create a div to contain the h1, p and img elements. Adjust the margins so that the image and text are cer
Hint: You horizontally center a div by giving it a width of 50% and a margin-left of 25%.
Hint: Set the image to have a width of 100% so it fills the div.
6. Read about the text-transform property on MDN docs to make the h1 uppercase with CSS.
https://developer.mozilla.org/en-US/docs/Web/CSS/text-transform
-->
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

6.4 Motivation Meme Project > solution > solution.html > html > body > div.poster > p

```html
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Motivation Meme</title>
  <link rel="stylesheet" href="./solution.css" />
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Libre+Baskerville&display=swap" rel="stylesheet">
</head>

<body>

  <div class="poster">
    <img class="motivation-img" src="./assets/images/daenerys.jpeg" src="daenerys with egg" />
    <h1>That Special Moment</h1>
    <p>When you find the perfect avocado at the supermarket</p>
  </div>
</body>

</html>
```

Code File Edit Selection View Go Run Terminal Window Help

Fri 10 Feb 12:45

solution.css — Web Development Projects

EXPLORER

WEB DEVELOPMENT PR...

- 2.4 Movie Rankin...
- 3.0 List Elements
- 3.1 Nesting and I...
- 3.2 Anchor Elem...
- 3.3 Image Eleme...
- 3.4 Birthday Invit...
- 4.0 File Paths
- 4.1 Webpages
- 4.3 HTML Porfoli...
- 5.1 Adding CSS
- 5.3 CSS Selectors
- 5.4 Color Vocab ...
- 6.0 CSS Colors
- 6.1 Font Properties
- 6.2 Inspecting C...
- 6.3 CSS Box Mo...
- 6.4 Motivation M...
  - assets
  - solution
    - assets
    - solution.css
    - solution.html
  - goal.png
  - index.html
  - style.css

index.html | solution.css

6.4 Motivation Meme Project > solution > solution.css > .poster

```css
body {
  background-color: black;
}
h1 {
  text-transform: uppercase;
  font-size: 3rem;
}

.poster {
  width: 50%;
  margin-left: 25%;
  margin-top: 100px;
  color: white;
  font-family: "Libre Baskerville", serif;
  text-align: center;
}

.motivation-img {
  border: 5px solid white;
  width: 100%;
}
```

Motivation Meme

http://127.0.0.1:3000/6.4 Motivation Meme Project/solution/solution.l



THAT SPECIAL MOMENT

When you find the perfect avocado at the supermarket